

Biochip Course

Electrochemical Detection

Analytical Techniques and Instrumentation

Marco Carminati

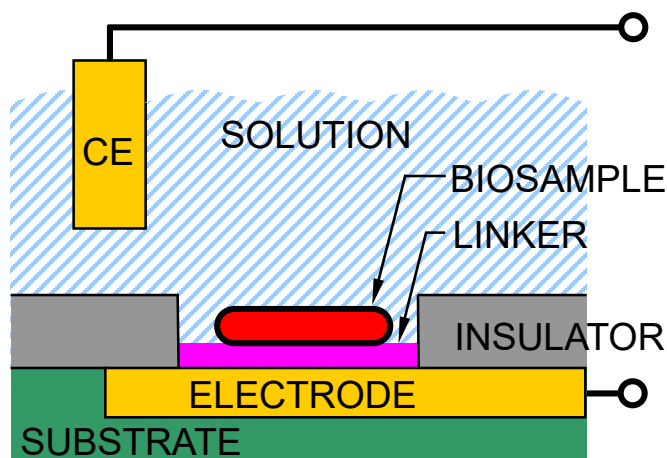
a.y. 2022/2023

March 20th 2023

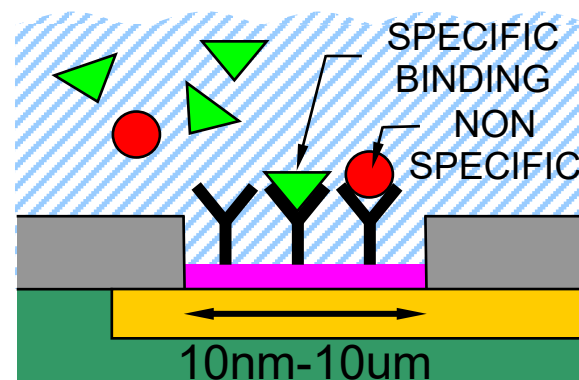


Sensing at the Electrochemical Interface

Use the ionic liquid as a top conformal conductive electrode



suitable for
affinity biosensors

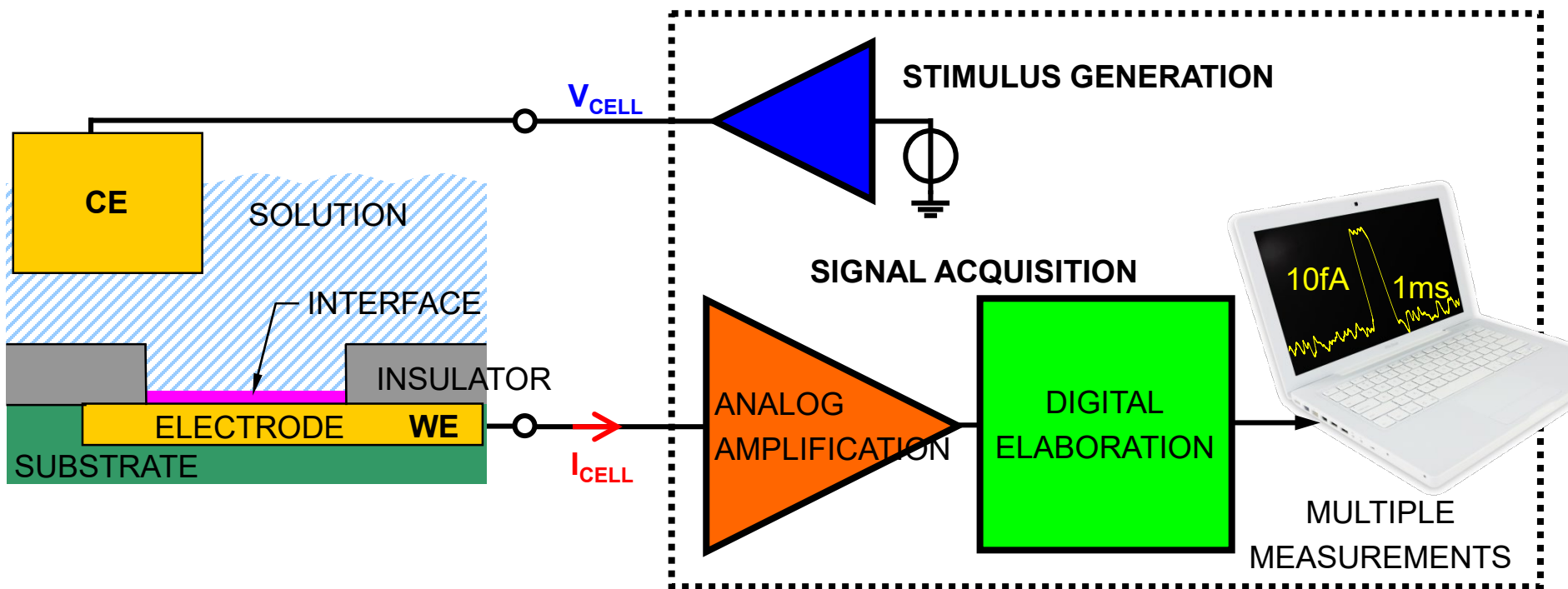


Probing charge transfer
and induction at the
interface affected by the
presence of the analyte



General Electrochemical Instrument

Potentiostatic configuration:



- Apply a voltage V_{CELL} and measure the current I_{CELL}



2 sections: **generation** and **sensing**

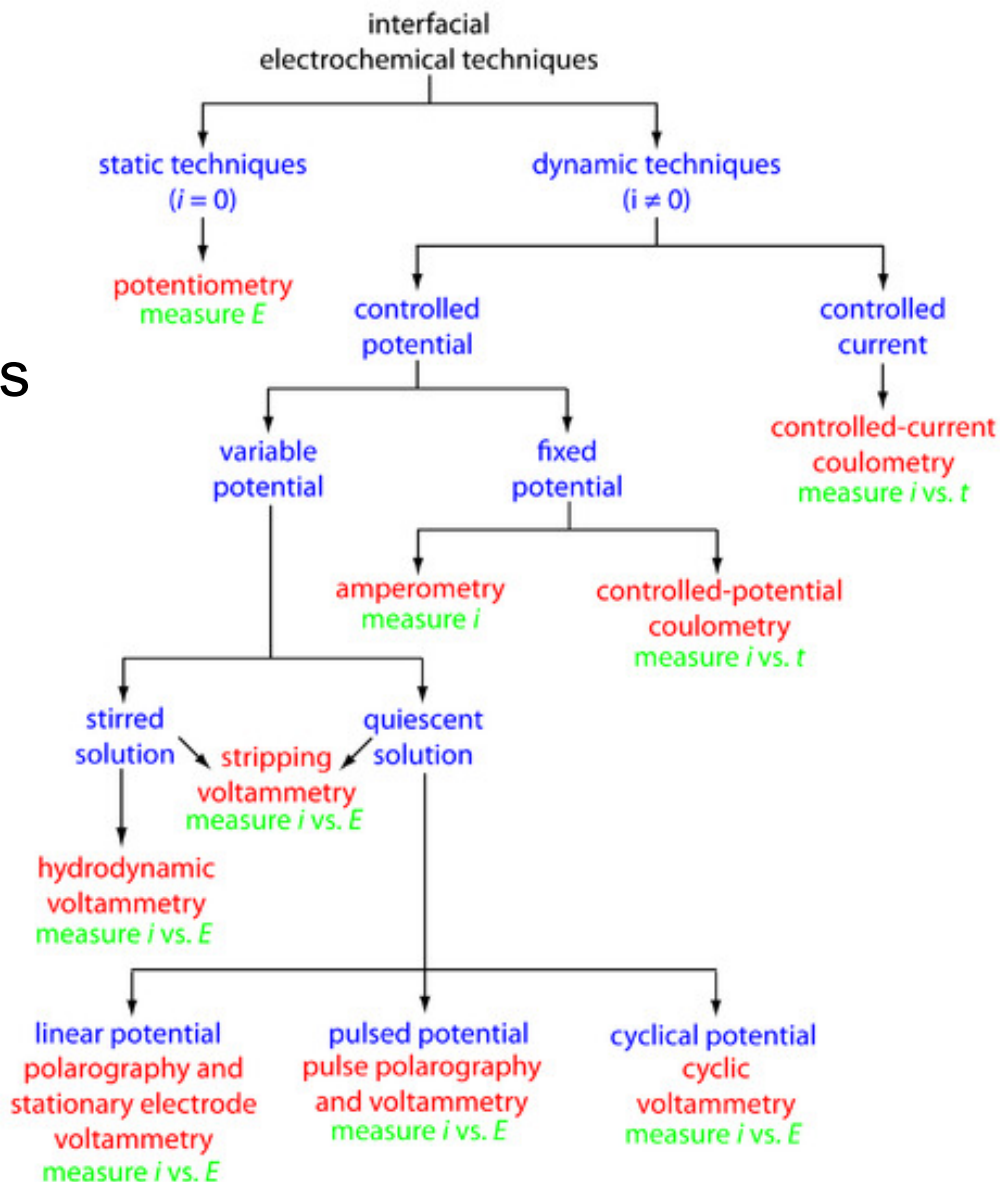
- Multiple measurement types by changing stimulation waveform



Families of Analytical Techniques

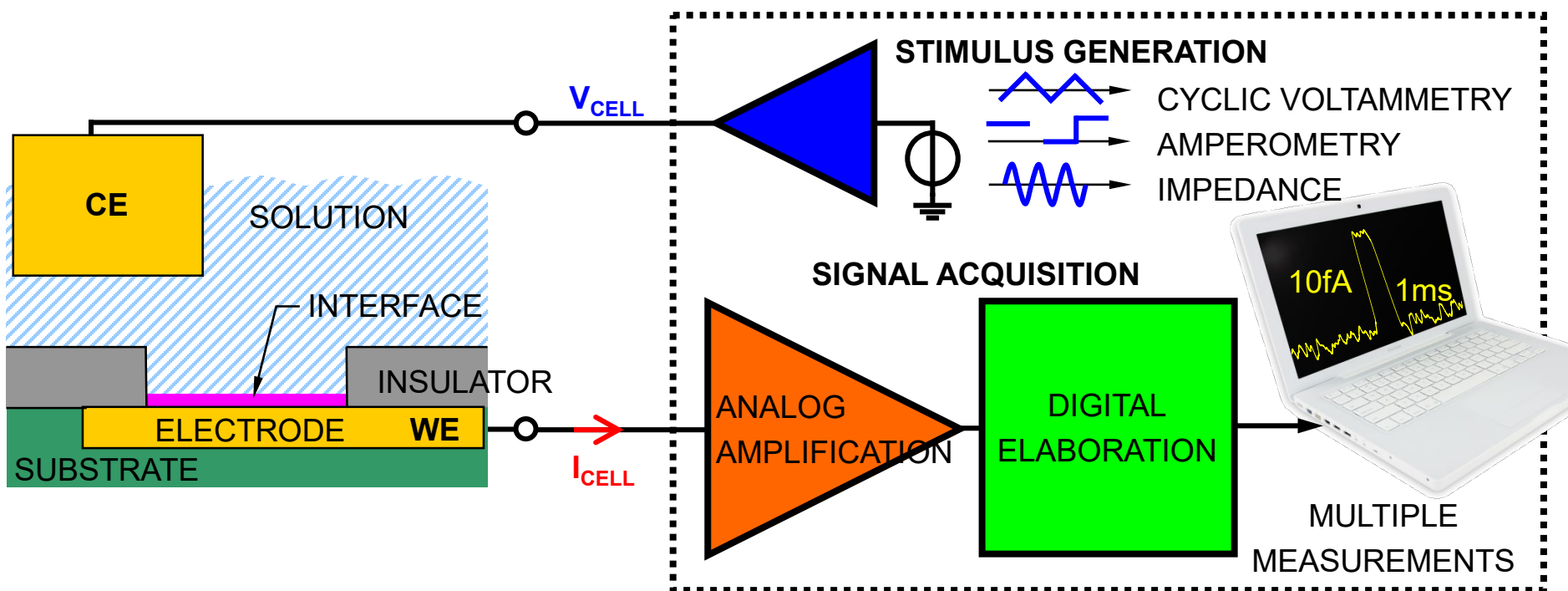
Detection techniques:

- Potential step/pulse methods
- Potential sweep methods
- Controlled-current techniques
- Methods based on impedance
- Scanning Electrochemical Microscopy





Families of Analytical Techniques

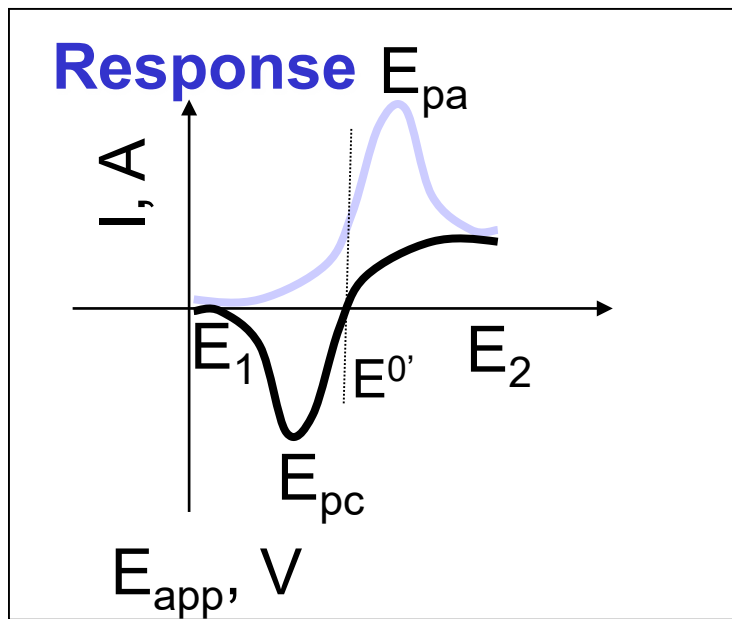
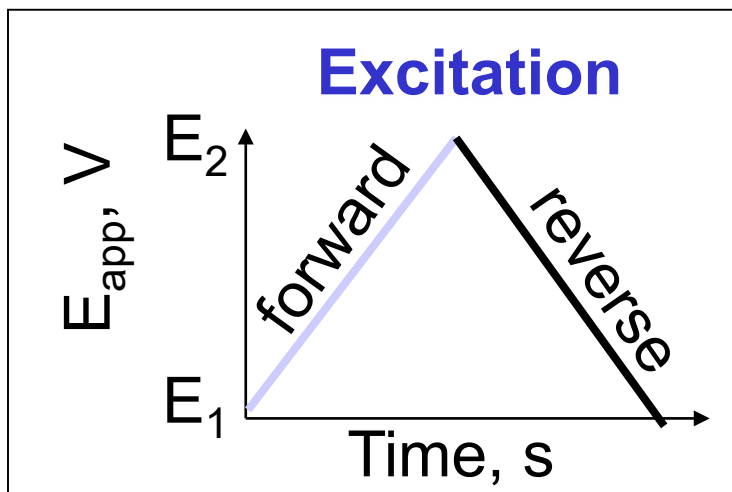


Major ones:

- *Cyclic Voltammetry*
- *Amperometry*
- *Impedance Spectroscopy and Tracking*



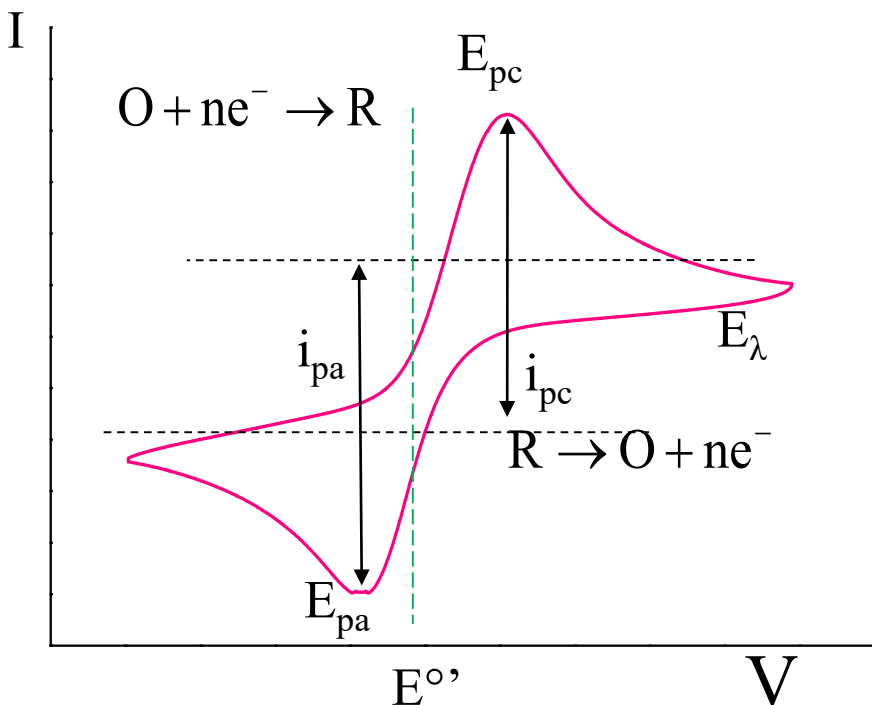
Cyclic Voltammetry



- Triangular wave: slope = **scan rate**
- **Important parameters:**
 - E_{pa} and E_{pc}
 - i_{pc} and i_{ac}
 - $E^0 = (E_{pa} + E_{pc})/2$
 - $\Delta E = |E_{pa} - E_{pc}|$
- E^0 formal redox potential
- detection of chemical reactions
- evaluation of electron transfer kinetics and diffusion rates



Peak Current (Reversible Systems)



$i_{pa} = i_{pc}$ for a reversible system

$$i_p = 0.4463 n F A C (D S R n q / k T)^{1/2}$$

n : n. of electrons

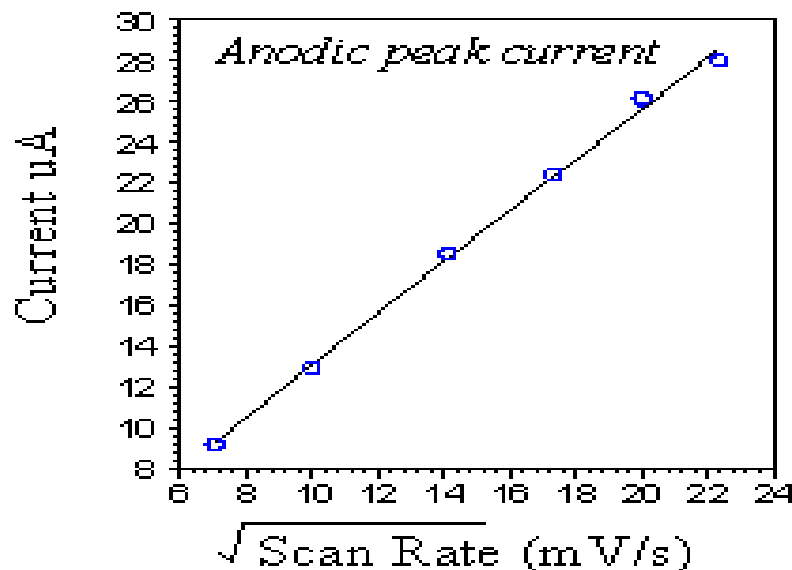
F : Faraday's constant (96485 C/mol)

A : electrode area;

C : concentration (mol/litre)

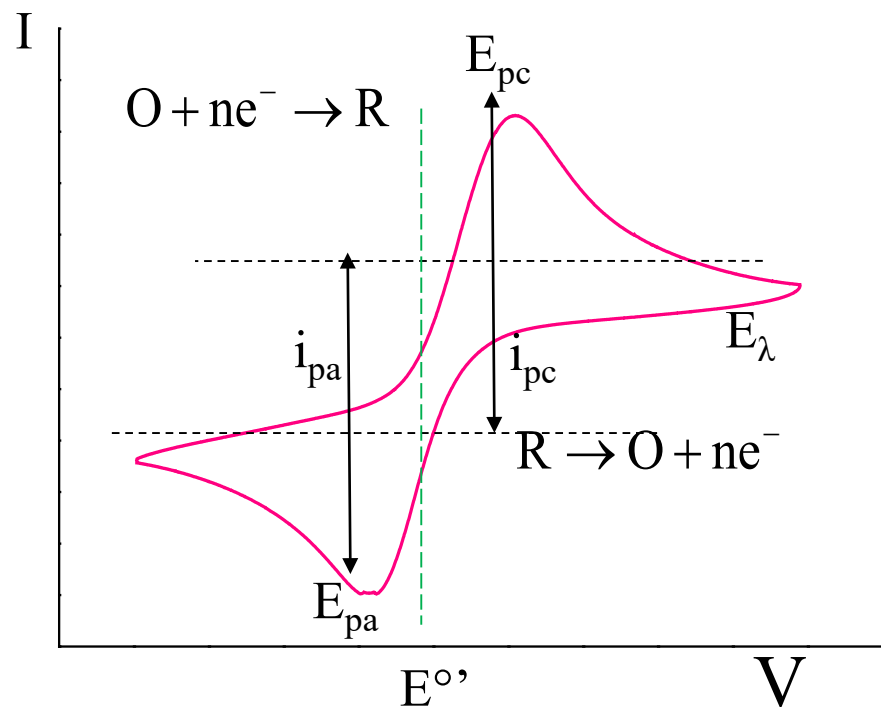
D : diffusion coefficient (cm^2/sec).

SR : scan rate



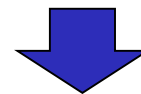


Peak Potential



$$\Delta E_p = |E_{pa} - E_{pc}| = 59/n \text{ mV at } 25^\circ\text{C}$$

independent of SR



depends on:
n and reversibility

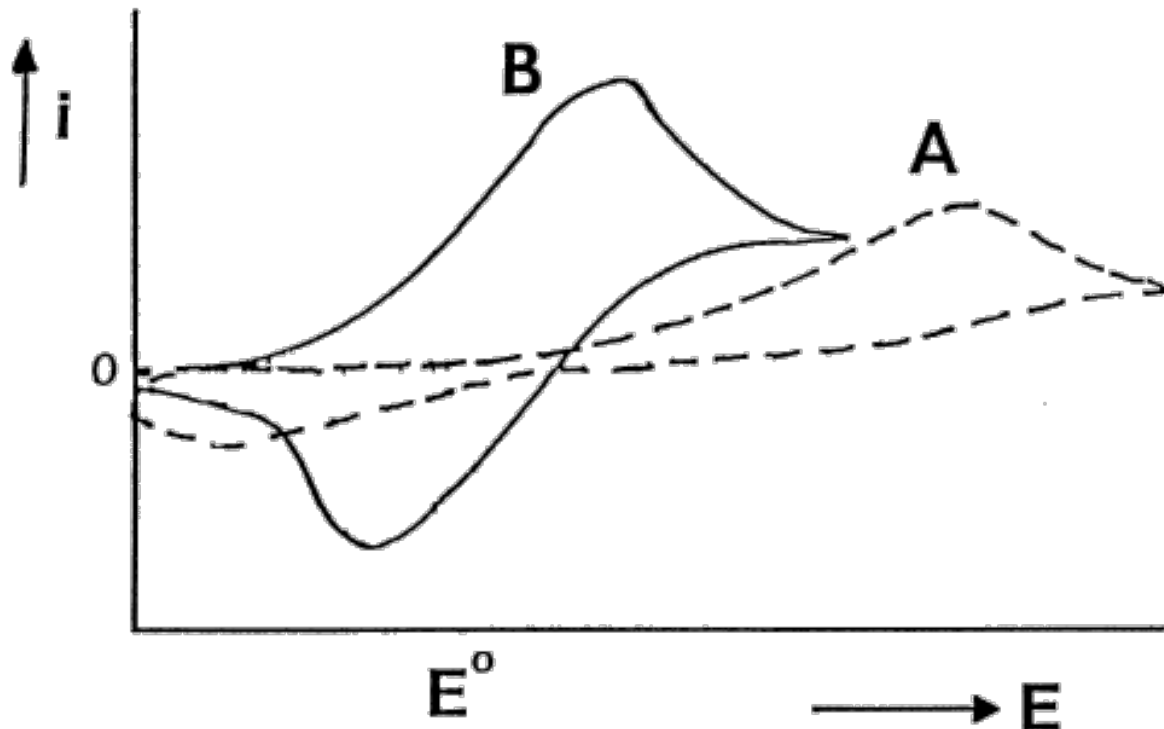
Chemical fingerprint

B - Quasi-reversible:

- $\Delta E_p > 59 \text{ mV}$ and ΔE_p increases with increasing SR

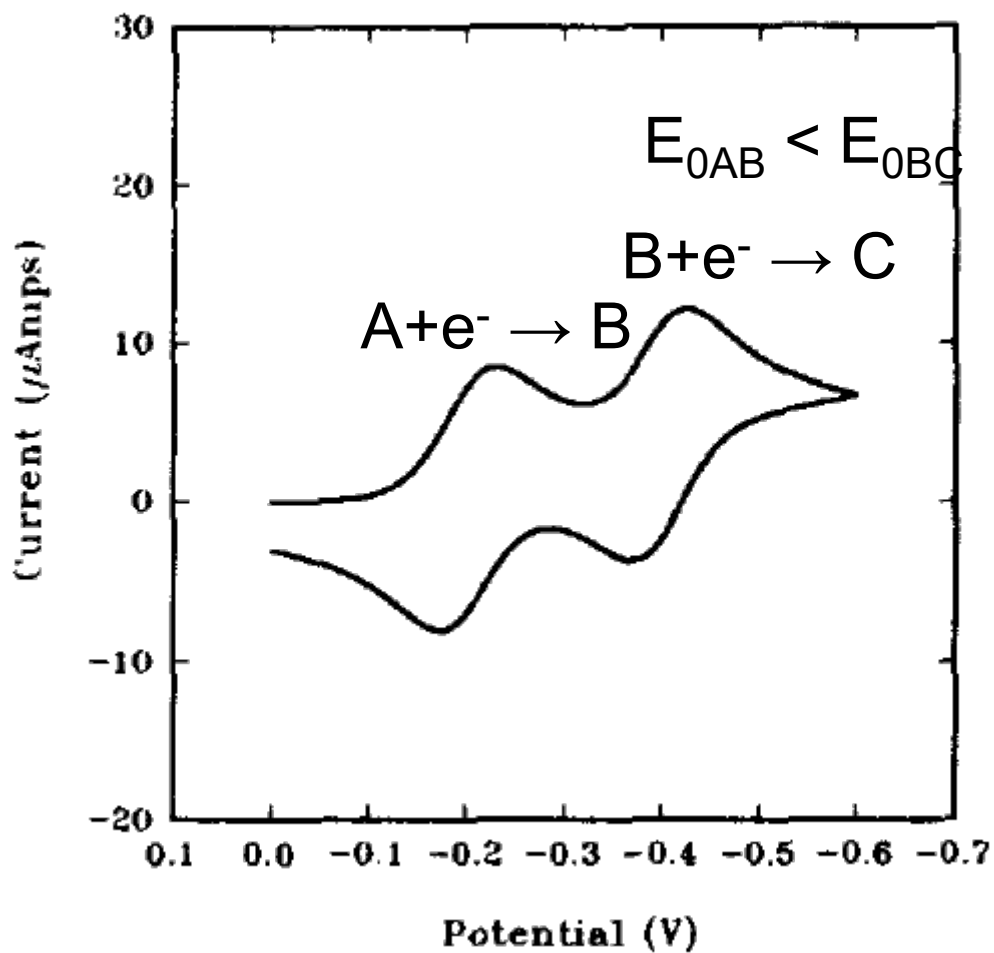
A - Irreversible:

- no return wave, 2 waves do not overlap

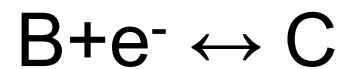




Multi-Electron Transfer



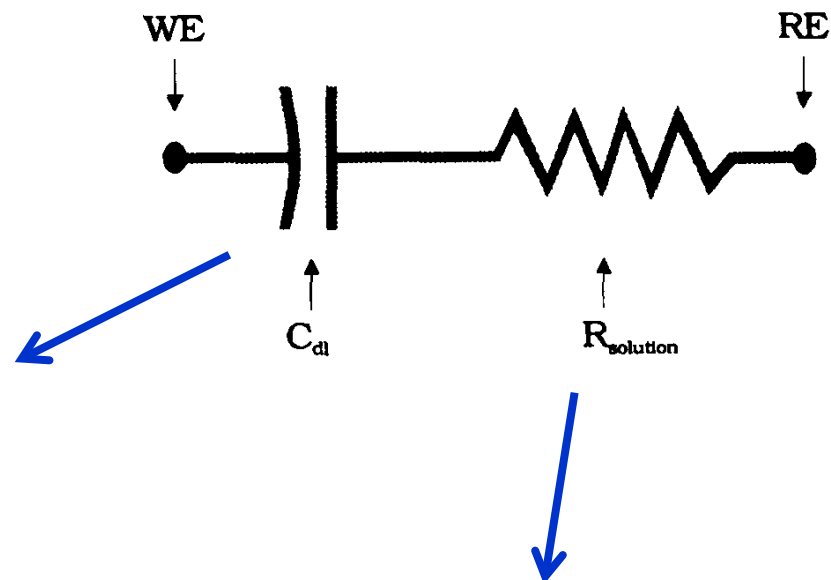
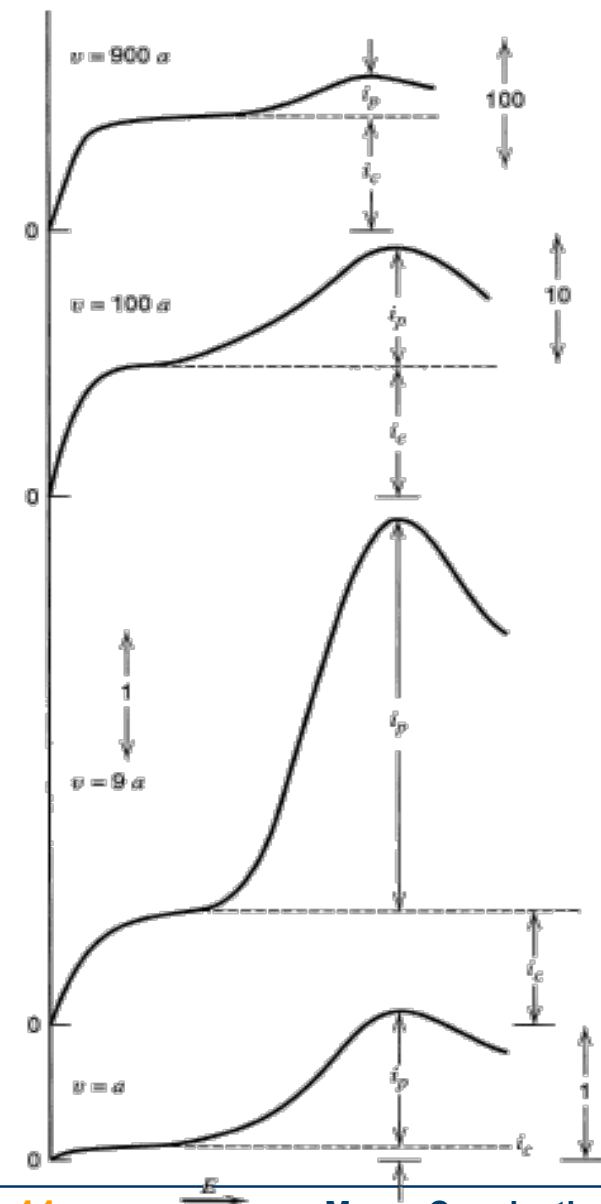
$$E_{0AB}$$



$$E_{0BC}$$



Spurious Effects



iR_{sol} drop

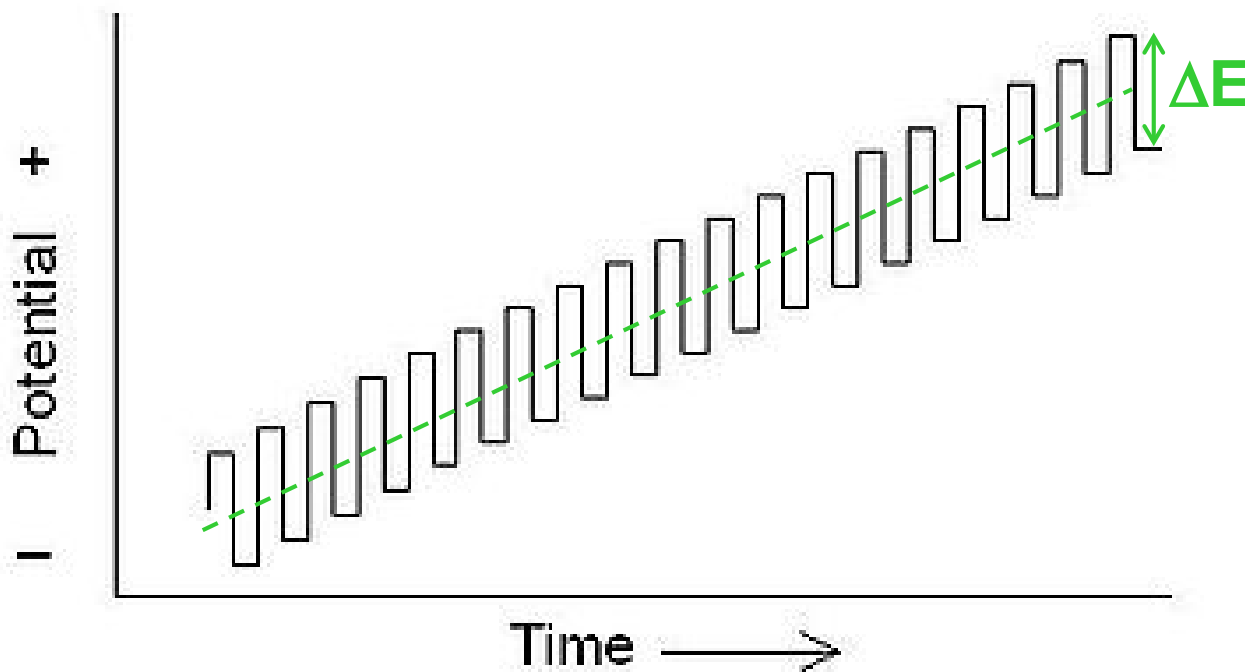
$C_{dl}R_{sol}$ time constant



Square Wave Voltammetry

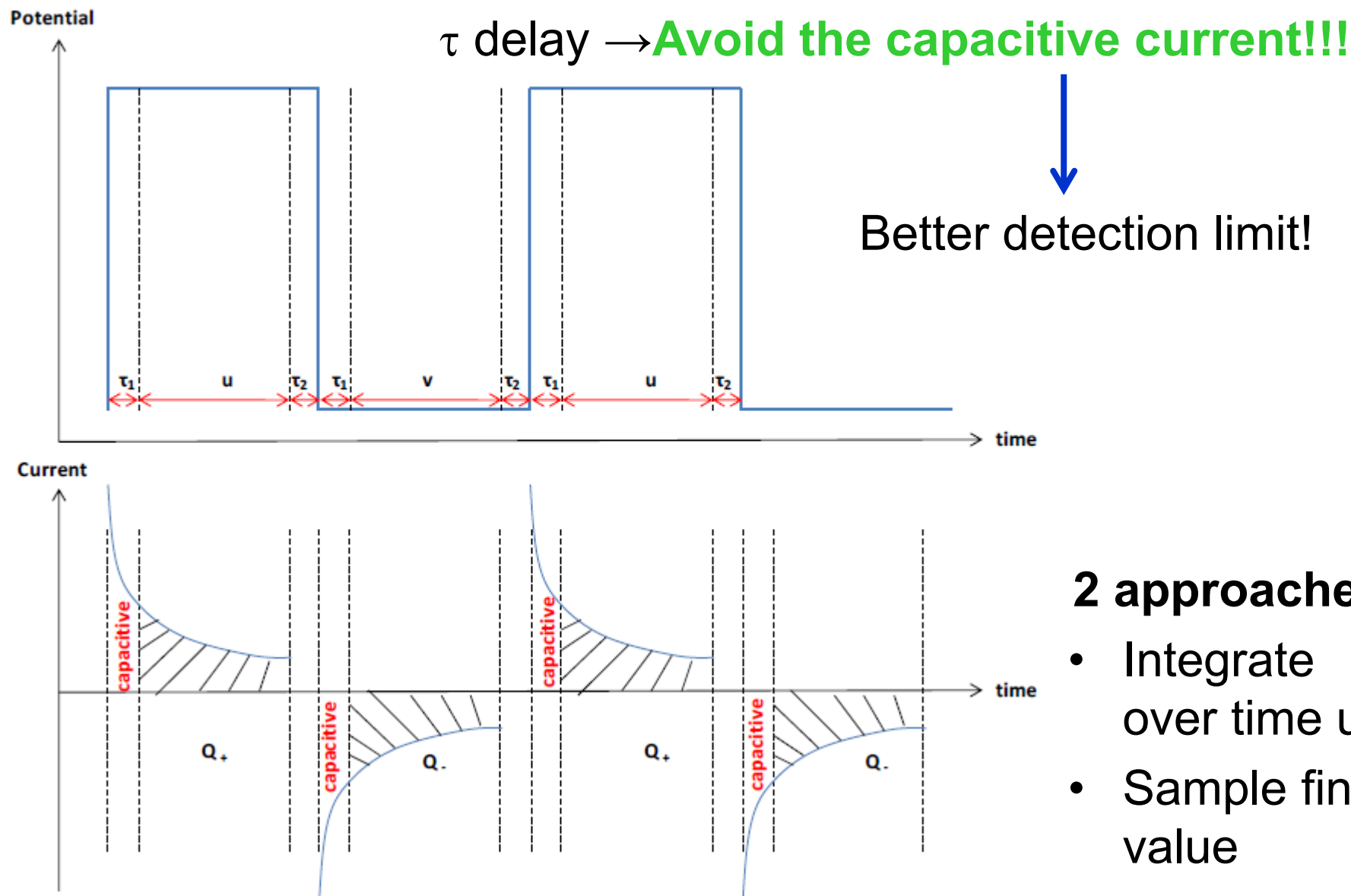
Evolution of linear sweep voltammetry: superposition of a **square wave** with large amplitude ΔE and take $i_{\max} - i_{\min}$

Obtain the *derivative* of the voltammogram (*differential square wave voltammetry*)





Square Wave Voltammetry

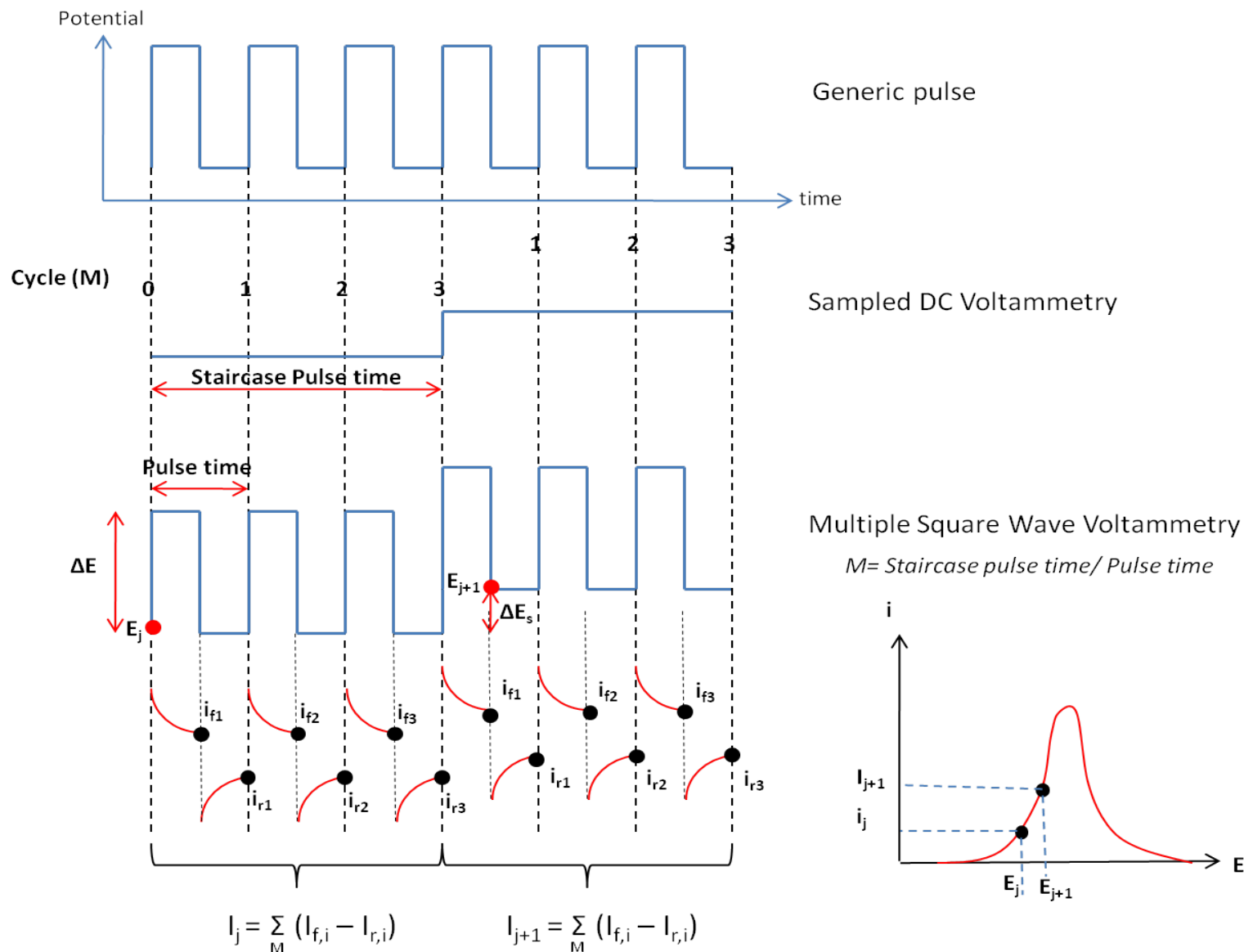


2 approaches:

- Integrate over time u v
- Sample final value



Multiple Square Wave Voltammetry

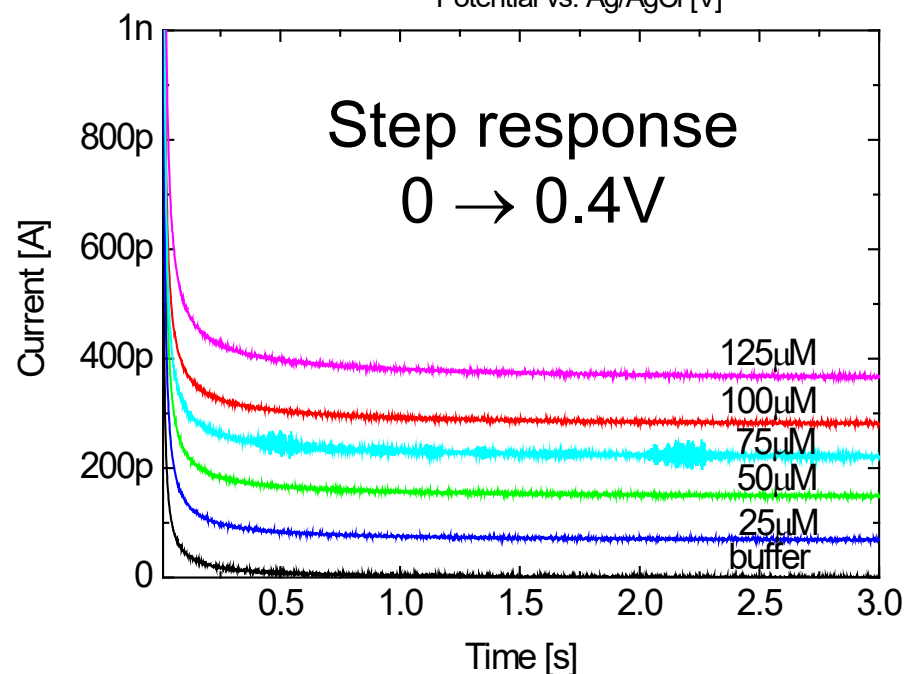
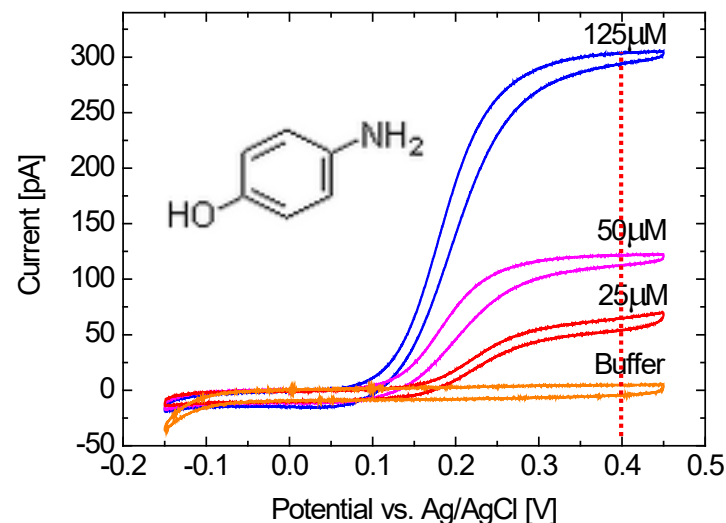
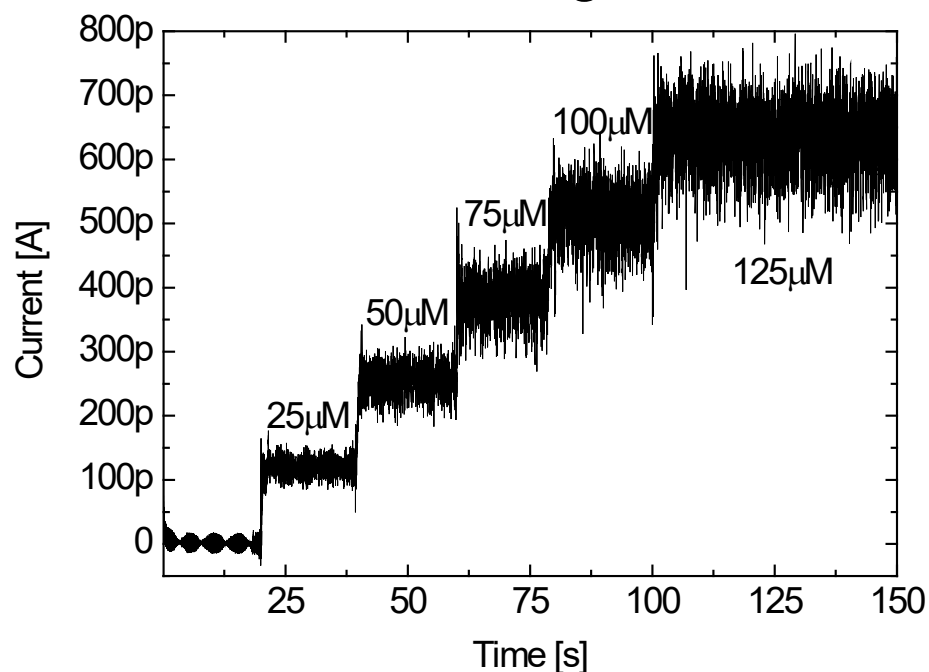




Recording current over time at a fixed WE potential

$$i \propto \text{concentration}$$

Fixed voltage 0.4V



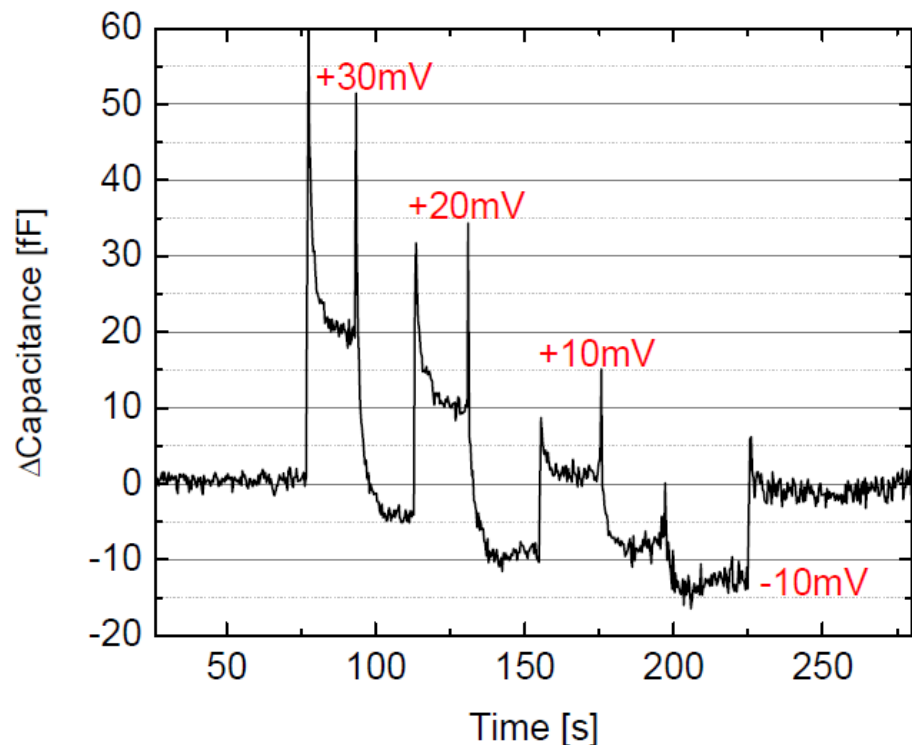


Electrochemical Impedance Sensing

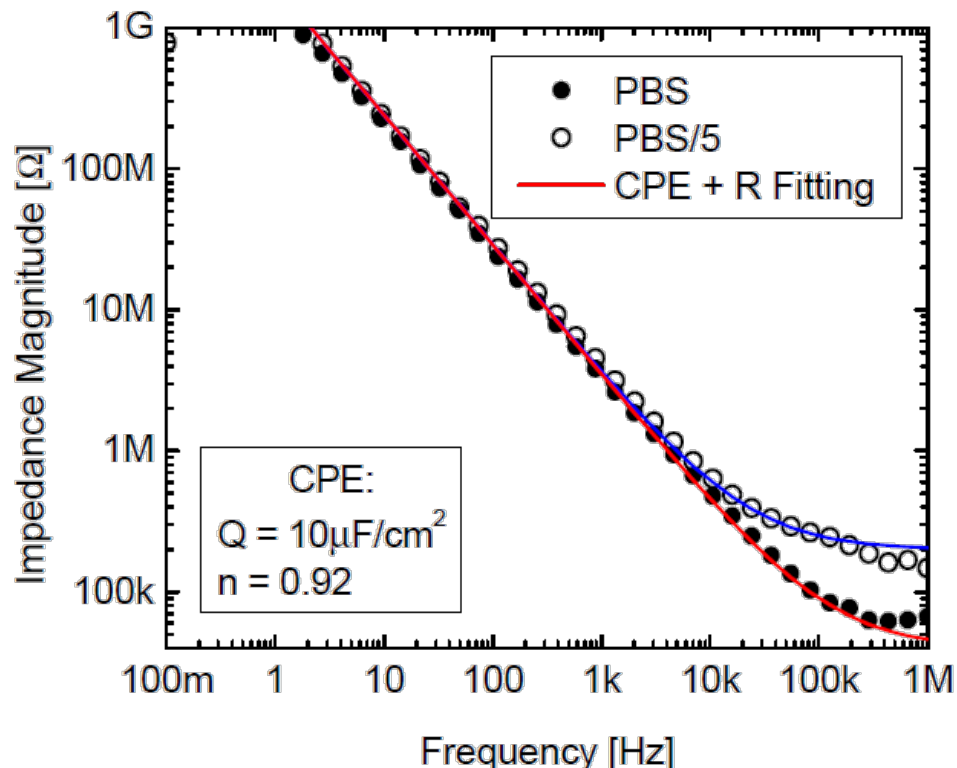
Sinusoidal excitation:

- Small signal (1-10mV) for linearity and minimal perturbation
- Wide frequency span: from <Hz to 100kHz-1MHz

Time tracking:



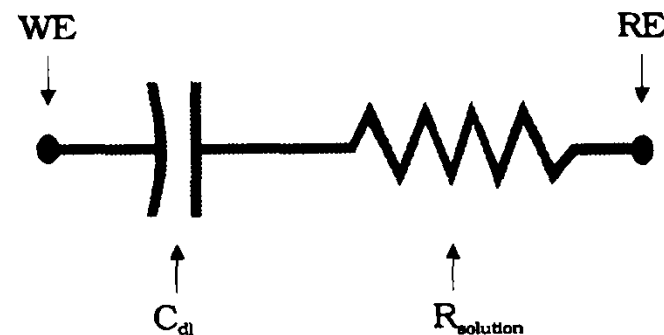
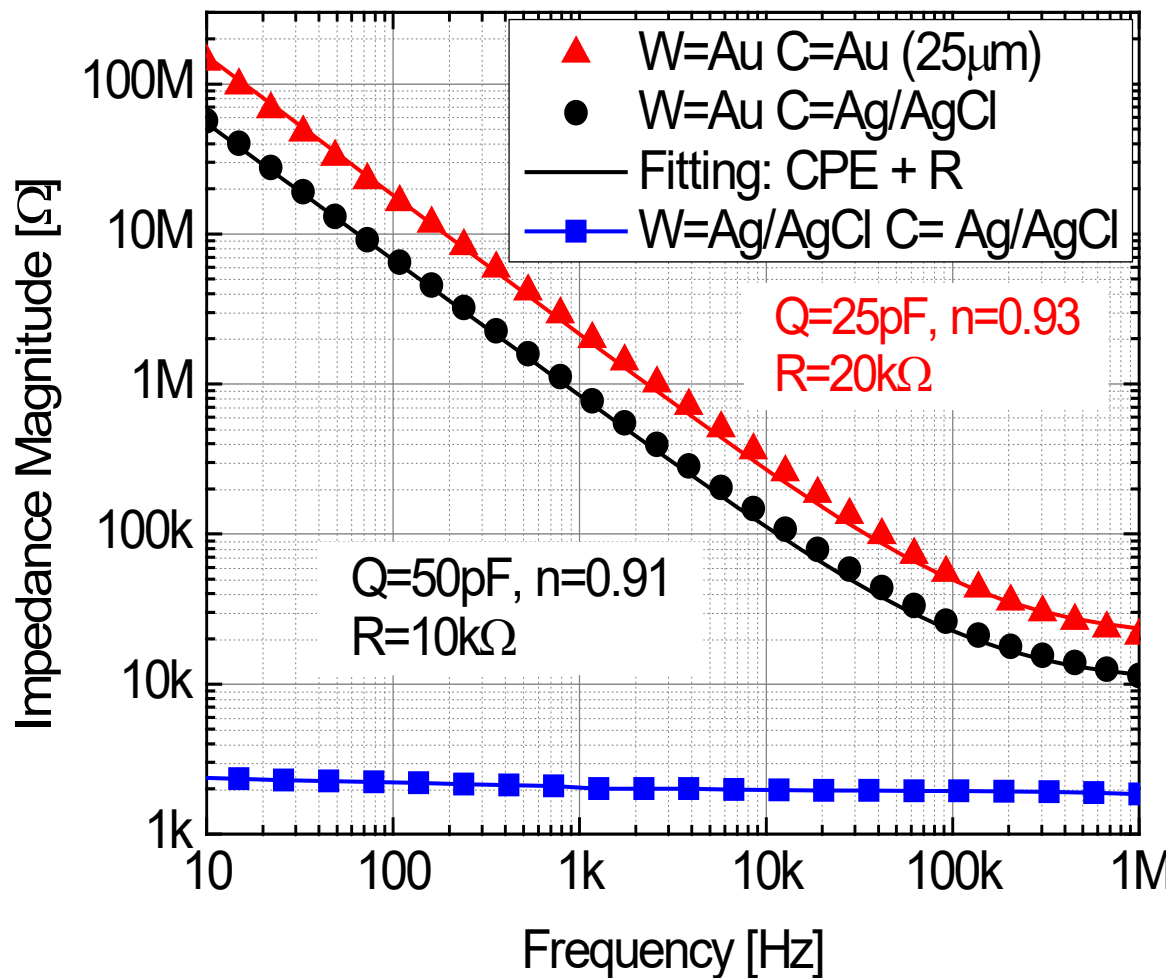
Frequency sweep:





Impedance Spectroscopy Application

Very sensitive to **interfacial properties**



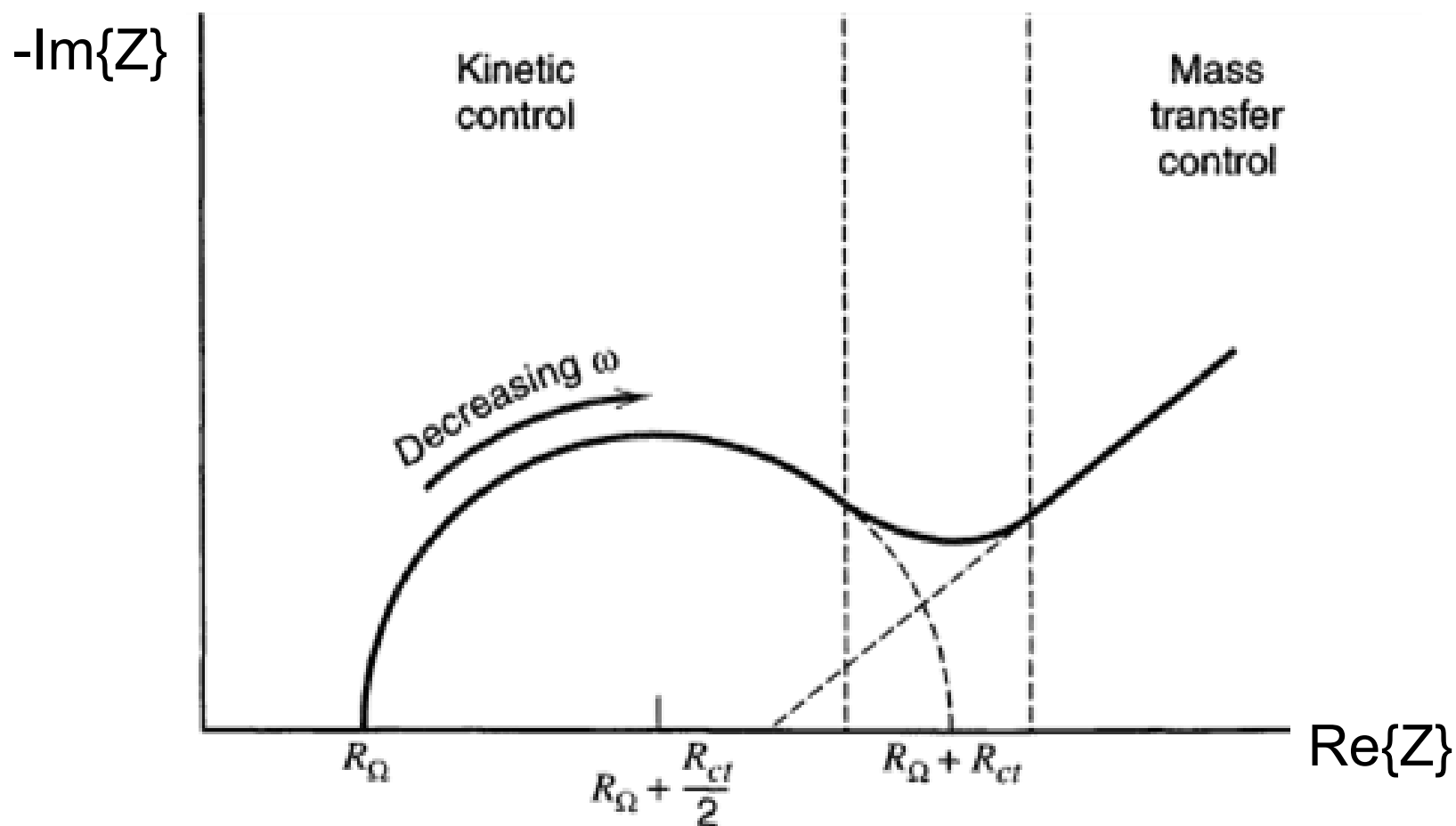
Applications:

- Coatings (paint)
- Corrosion
- Surface roughness
- Affinity biosensors
- Dielectric properties
- Battery SoC

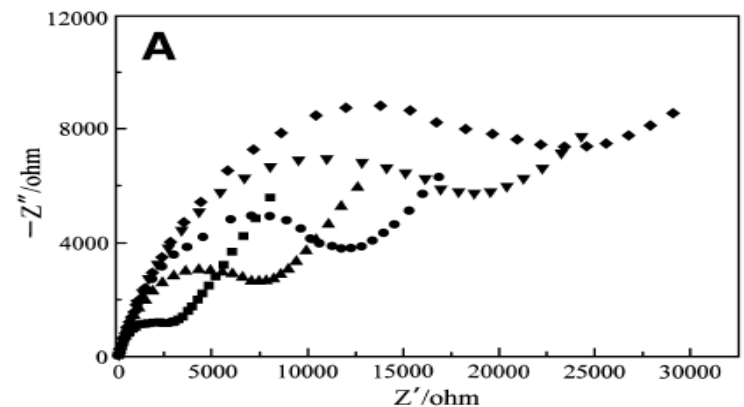
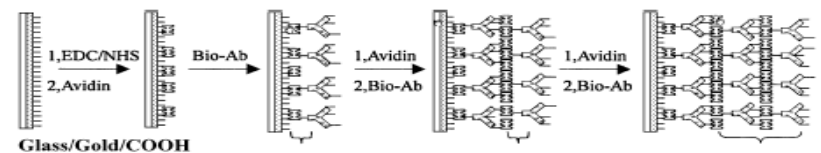
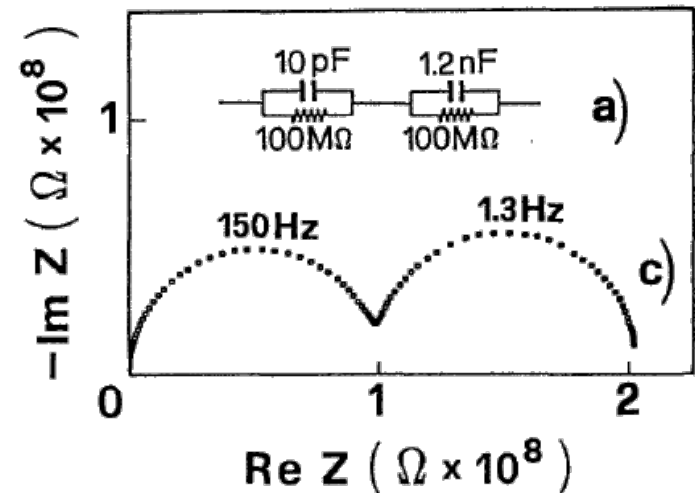
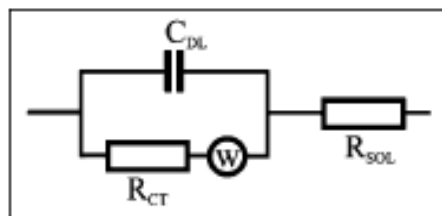
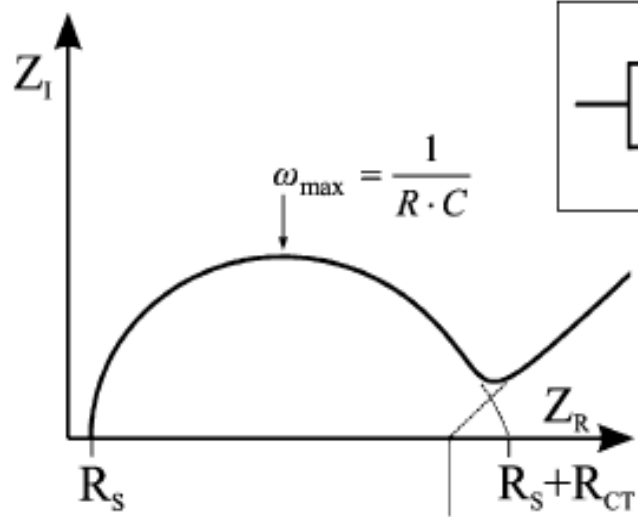
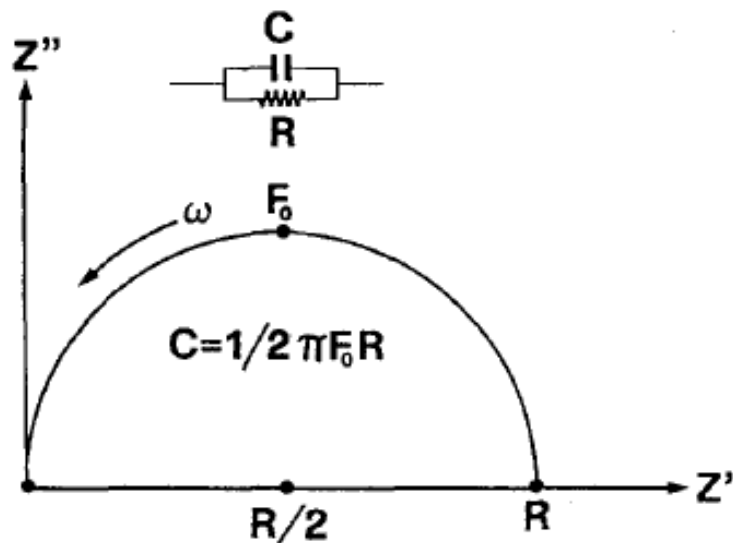


The Cole-Cole Plot

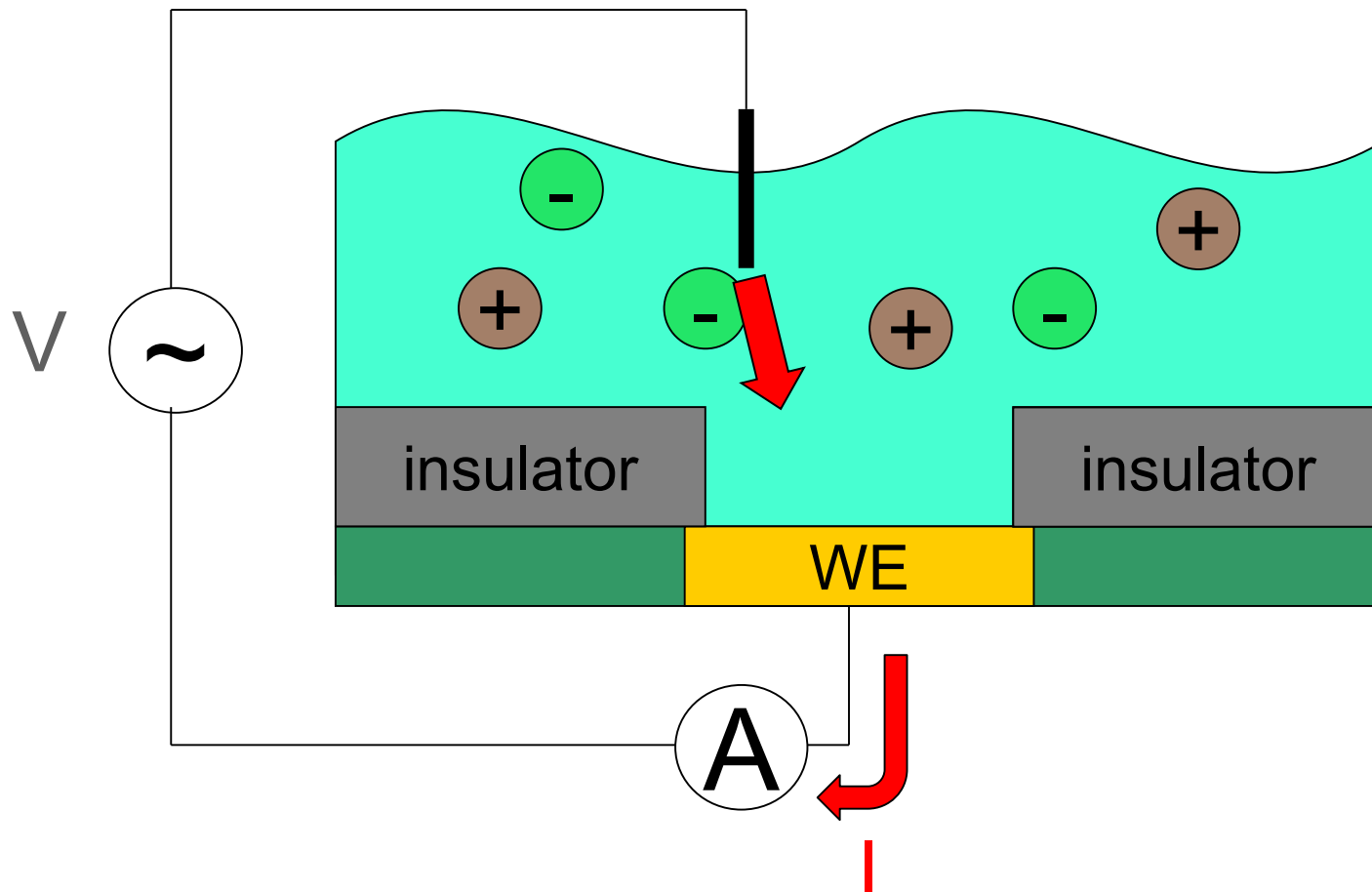
EIS spectra visualization: a very common representation in Electrochemistry, alternative to Bode (Nyquist diagram)



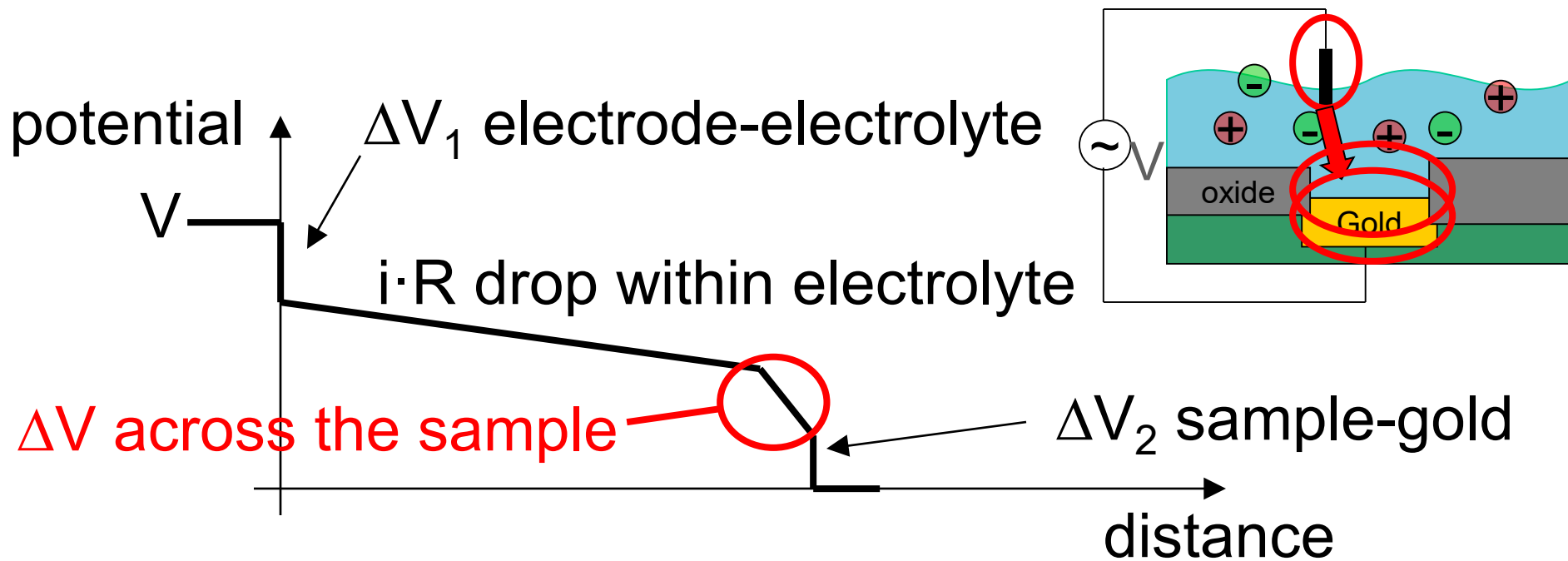
Cole-Cole Parameter Extraction



Two-Electrode Measurement



Two-Electrode Measurement



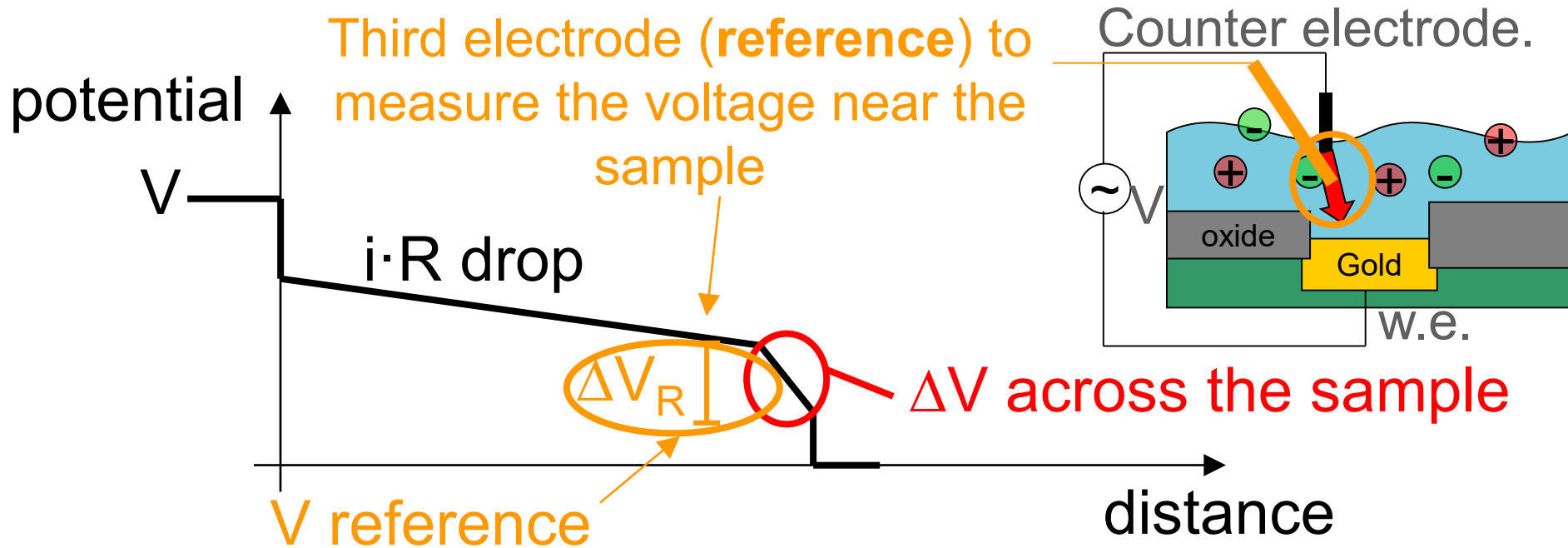
$\Delta V_1, i \cdot R$ and ΔV_2 change with current

ΔV across the sample is current-depend

→ not well controlled



Three-Electrodes Setup

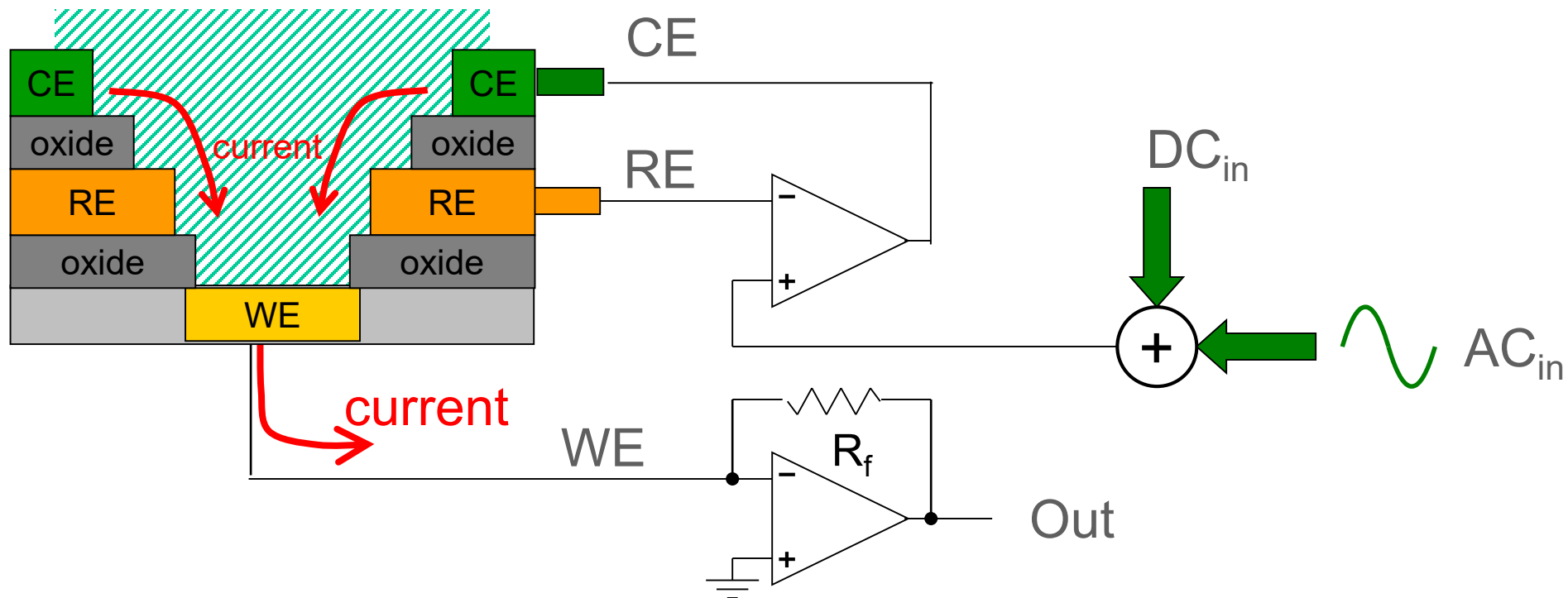


- Avoids internal polarization of reference electrode
- ΔV_R constant: **no current**; silver-silver chloride, calomel...
- reference electrode near the sample to compensate for major portion of cell iR drop



A New Instrument: the Potentiostat

Combining current detection and voltage-control loop





Purpose: provide **stable potential** against which other potentials can be reliably measured

Criteria:

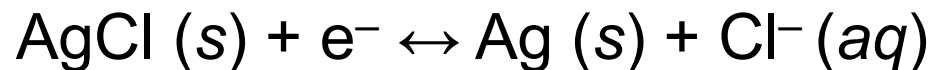
- stable in potential (time, temperature)
- reversible
- reproducible
- potential shouldn't be altered by passage of small current = not polarizable
- easy fabrication and handling



The Silver/Silver Chloride Electrode

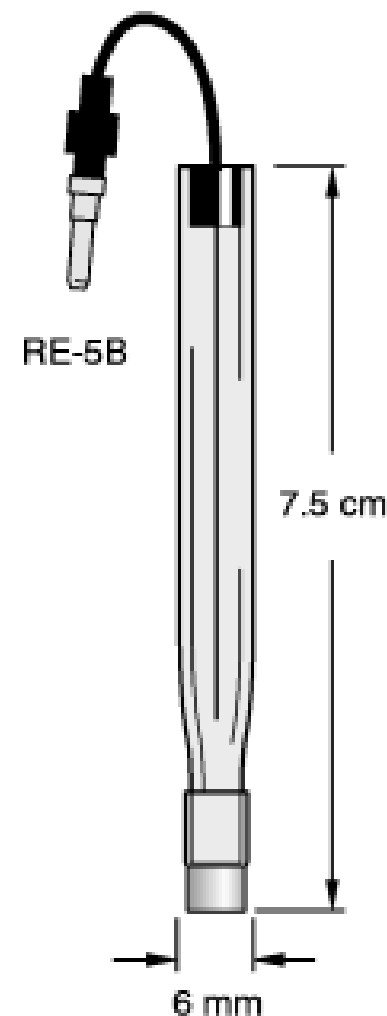
Ag wire coated with AgCl(s), immersed in NaCl or KCl solution separated by a ion-permeable membrane

It is stable in liquid that has large quantity of Cl⁻ such as the **biological fluid**.



$$E^\circ = +0.222 \text{ V}$$

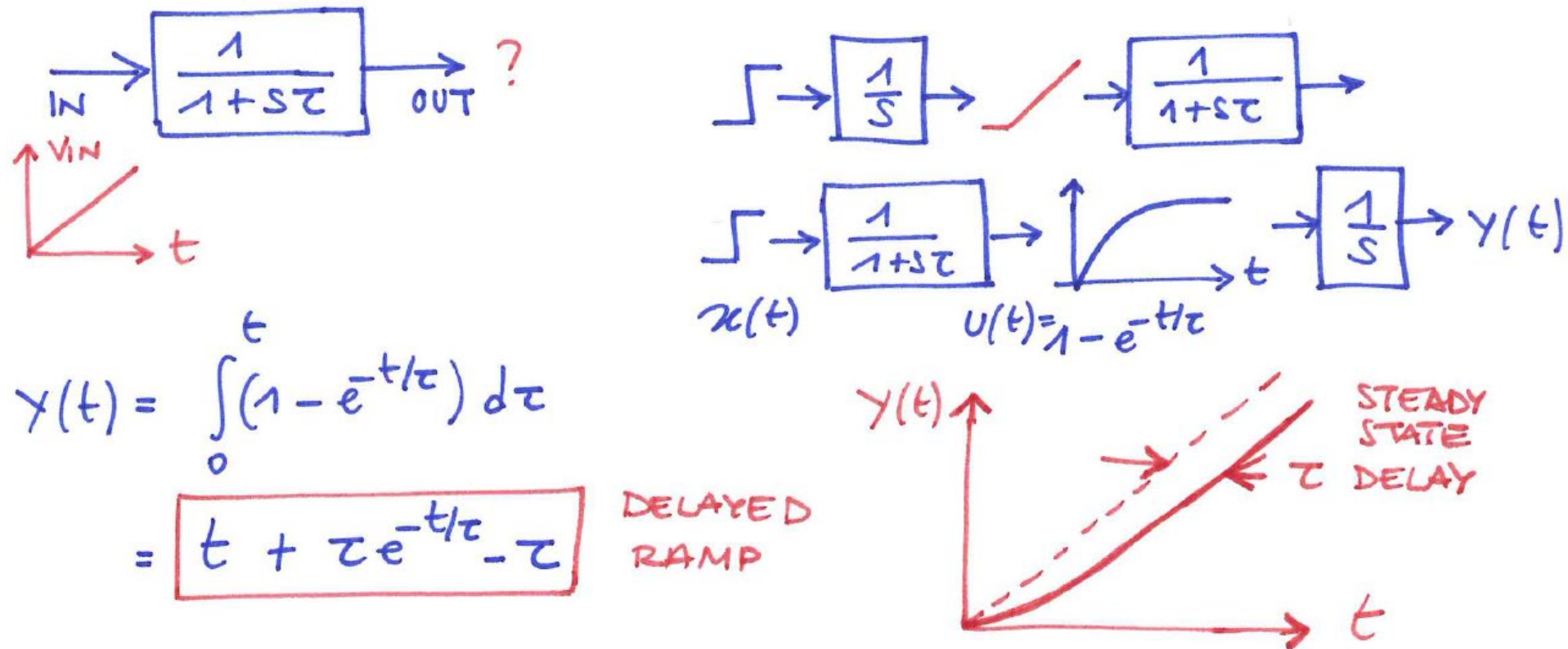
Stable up to +275°C





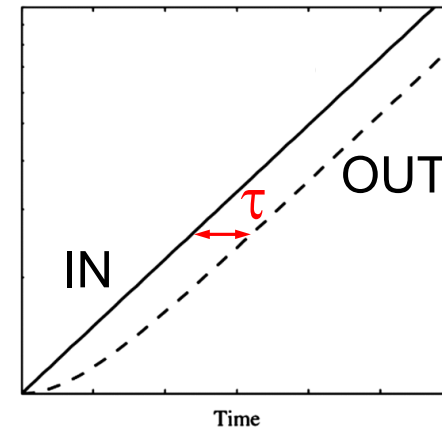
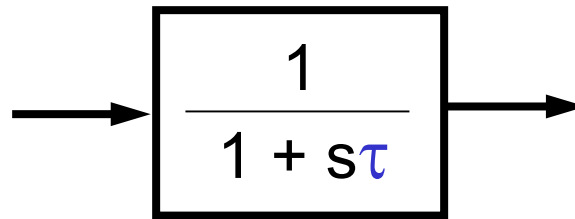
Single Pole Response to a CV Ramp

In order to design the circuit for CV measurement, the response of a 1st order system to a ramp input waveform is needed:



Generation (cyclic voltammetry):

- Accuracy: $\sim 1\text{mV}$ (vs. a 59mV characteristic feature)
- Bandwidth: $\text{BW [Hz]} = 40 \cdot \text{SR [V/s]}$



$$\Delta V_{\text{error}} = \tau \cdot \text{SR} = 4\text{mV} \rightarrow 1/(2\pi \cdot \tau) = \text{SR}/(2\pi \cdot 4\text{mV})$$

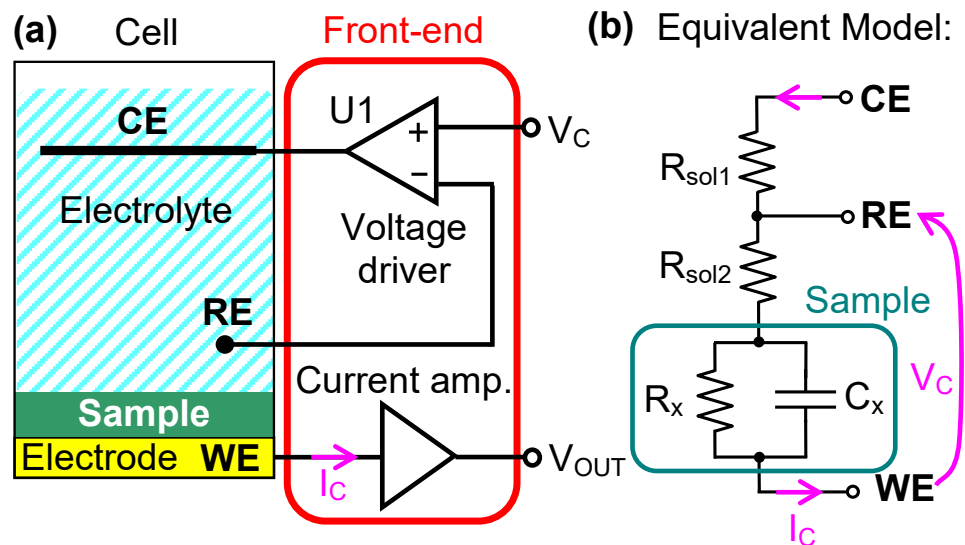
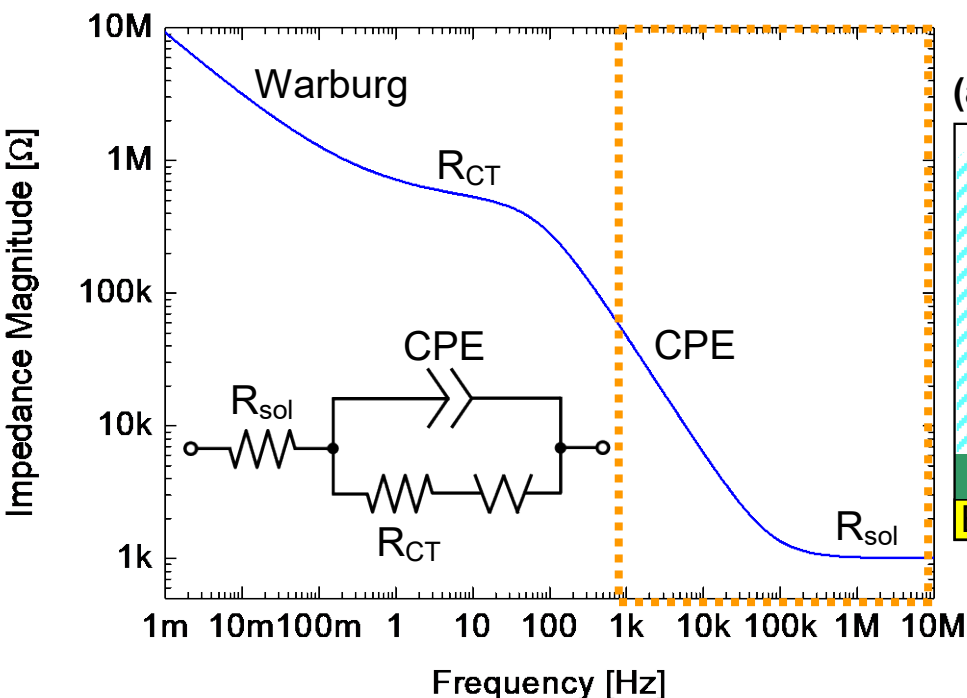
- Capacitive load: depends on the electrode area (C_{DL})

Sensing:

- Sensitivity: scales with electrode area
- Bandwidth: adequate for impedance and voltammetry



Design Approach



- $R_{sol} = \rho/4r$ (66Ωcm PBS)
- R_x = charge transfer
- $C_x = 0.1\text{pF}/\mu\text{m}^2$ (PBS)

Modulated by the sample

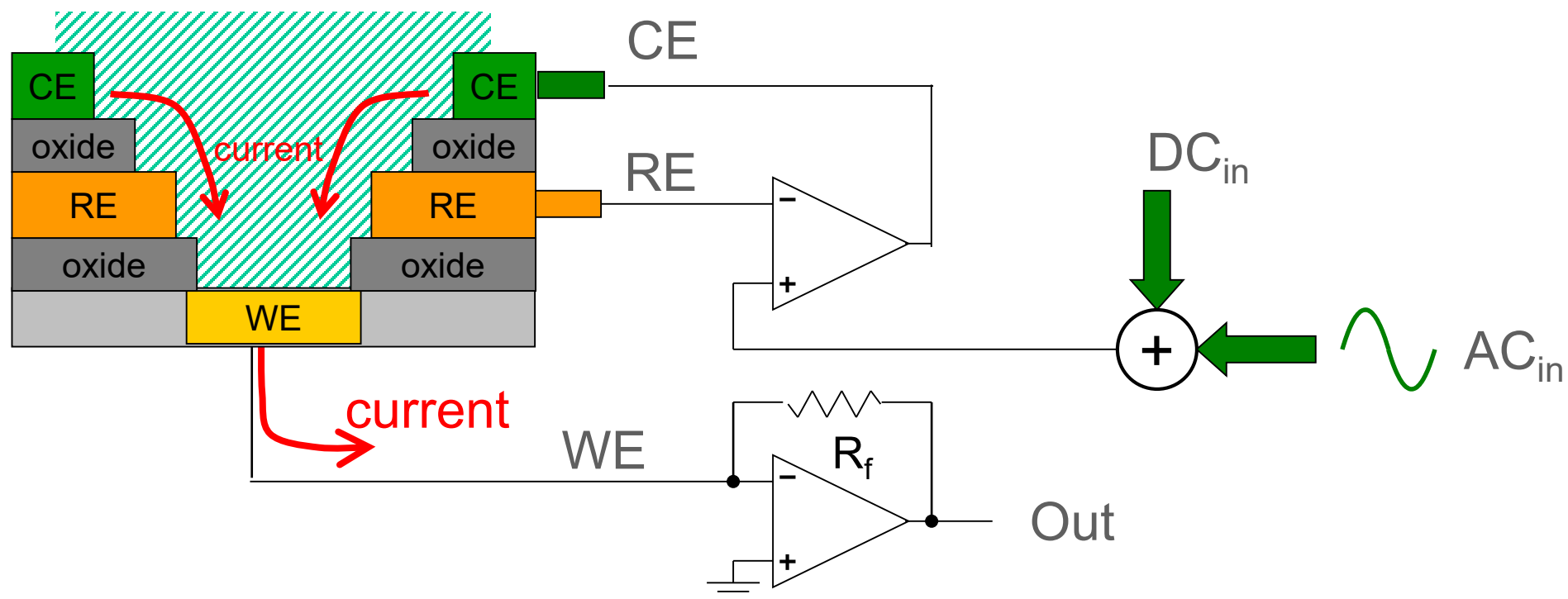
A **simplified model** of the interface can be adopted for design purposes (stability)

+ stray capacitances



Design Approach

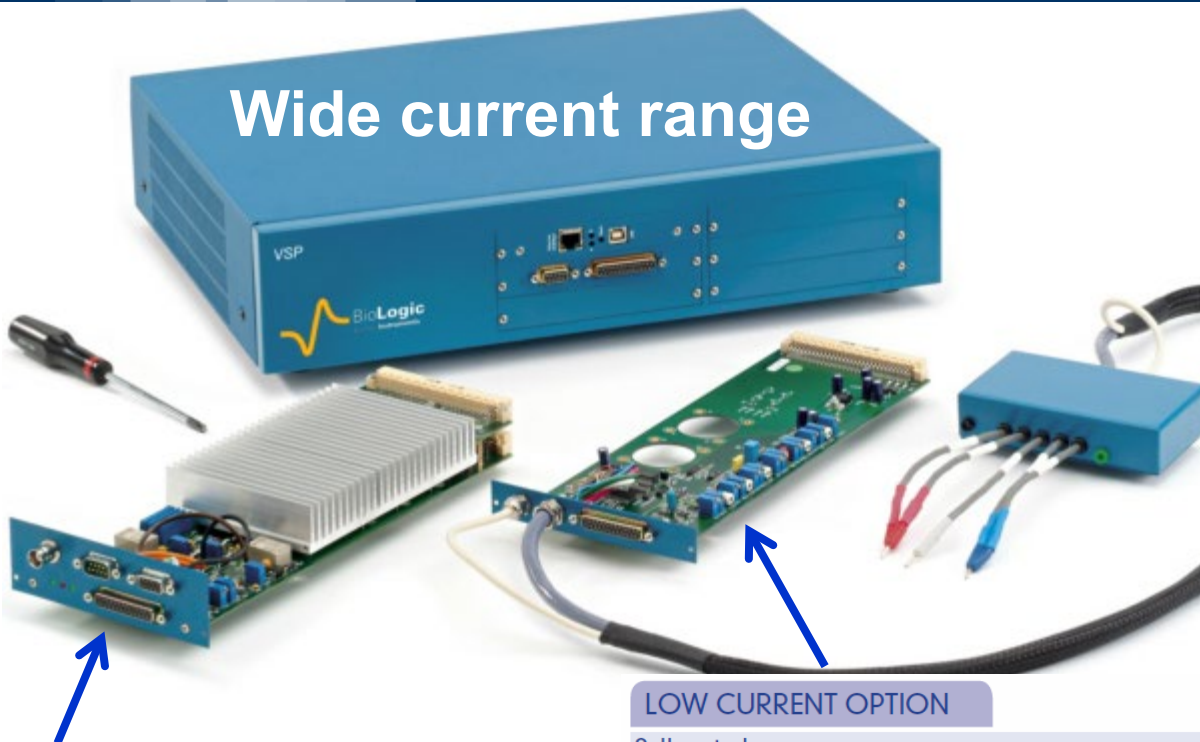
- Potentiostatic loop must be stable (electrode area)
- Adder to combine DC and AC stimuli
- R_f sized depending on the input current range (and BW)
- Opamp virtual ground can be biased to a desired potential





Commercial Workstations

Wide current range



SPECIFICATIONS

Cell control

Connection	2, 3, 4 or 5 terminals (+ ground)
Compliance	20 V adjustable from ± 10 V to 0-20 V
Maximum current	± 400 mA continuous
Maximum potential resolution	300 μ V on 20 V programmable down to 5 μ V on 200 mV
Maximum current resolution	0.004% of the dynamic range
Maximum current resolution	760 pA on the 10 μ A range
Accuracy (DC)	< 0.1% FSR*
Rise time	(10% - 90%) < 2 μ s (No load)
Acquisition time	20 μ s

Current measurement

Ranges	Automatic on every range ± 10 μ A to ± 400 mA (7 ranges)
Maximum resolution	0.004% of the range
Acquisition speed	200000 samples/second
Accuracy (DC)	< 0.1% FSR*

Potential measurement

Ranges	± 2.5 V, ± 5 V, ± 10 V, ± 10 V adjustable
Maximum resolution	0.0015% FSR*, down to 75 μ V
Acquisition speed	200000 samples/second
Accuracy (DC)	< 0.1% FSR*

Electrometer

Inputs	3 potential measurements
Impedance	> 10^{12} ohms in parallel with < 20 pF
Bias current	< 5 pA

EIS OPTION

Impedance

Frequency range	10 μ Hz to 1 MHz
Amplitude	1 mVpp to 1 Vpp 0.1% to 50% of the current range

General

Dimensions	435 x 335 x 95 mm
Weight	8.0 kg
Power	85-264 V, 47-440 Hz
PC configuration	Pentium IV, Windows 2000, XP or Vista

LOW CURRENT OPTION

Cell control

Maximum current	± 100 mA continuous
Maximum current resolution	0.004% of the dynamic range, program
Applied current accuracy	76 fA on the 1 nA range < 1% FSR* on the 1 nA range < 0.5% FSR* on the 10 nA range < 0.1% FSR* on the other ranges

Current measurement

Ranges	± 1 nA, ± 10 nA, ± 100 nA, ± 1 μ A
Maximum resolution	0.004% of the range down to 76.3 fA
Accuracy	< 1% FSR* on the 1 nA range < 0.5% FSR* on the 10 nA range < 0.1% FSR* on the other ranges

Electrometer

Impedance	10^{14} ohms in parallel with 1 pF
Bias current	60 fA typical, 150 fA max at 25 $^{\circ}$ C
Bandwidth	1 MHz

4 A BOOSTER KIT

Cell control

Maximum current	± 4 A continuous
Potential ranges	± 10 V at 4 A
Rise and fall time	Potentiometric mode: 15 μ s
10% to 90%	Galvanic mode: 100 μ s

Measurement

Potential accuracy (DC)	< 0.1% FSR*
Current accuracy (DC)	< 0.2% FSR*
Current noise	1 mA peak to peak (0-100 kHz) at 4 A
Potential noise	0.6 mV peak to peak (0-100 kHz)

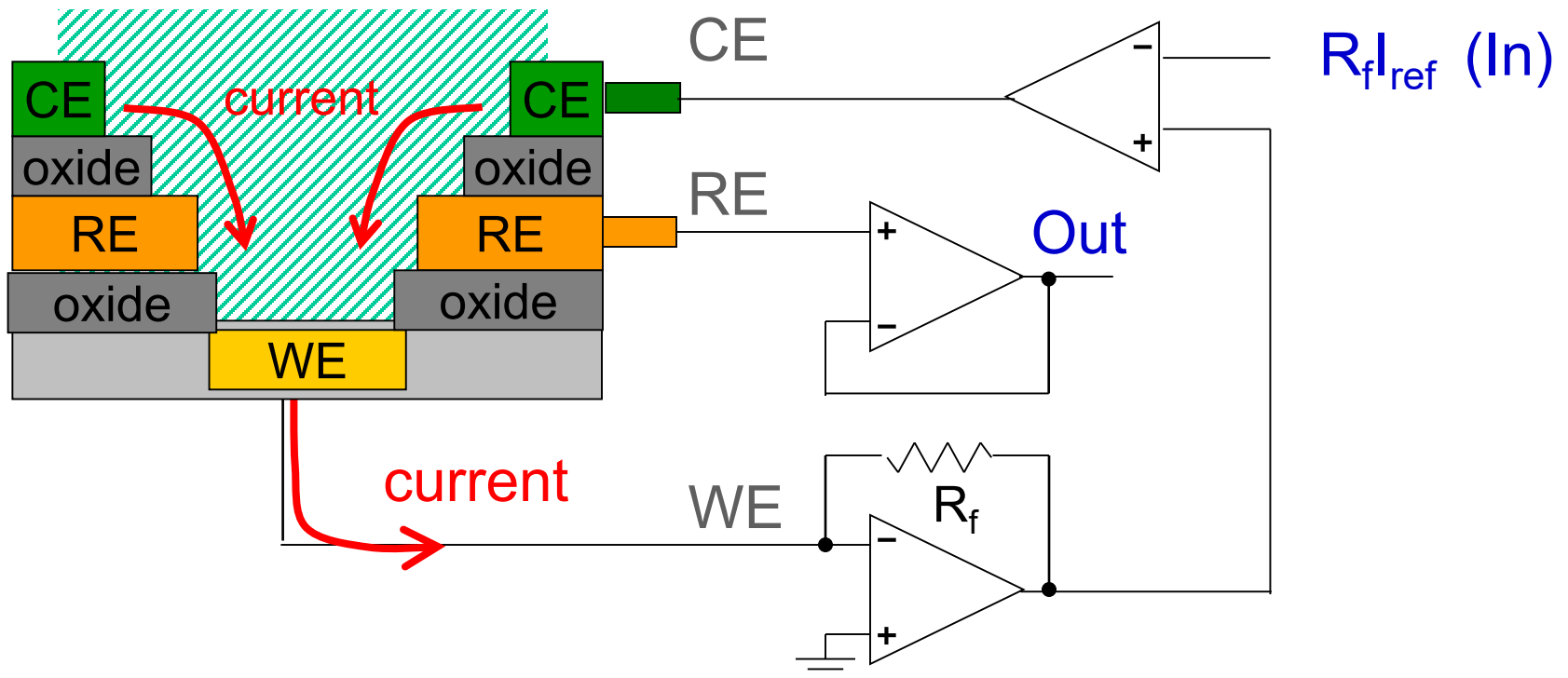
Electrometer

Impedance	10^{10} ohms
Inputs	3 potential leads with 2 differential voltages
Bandwidth	1 MHz



Galvanostatic Regulation

Feedback loop to control the current (desired set-point) and measure the voltage:

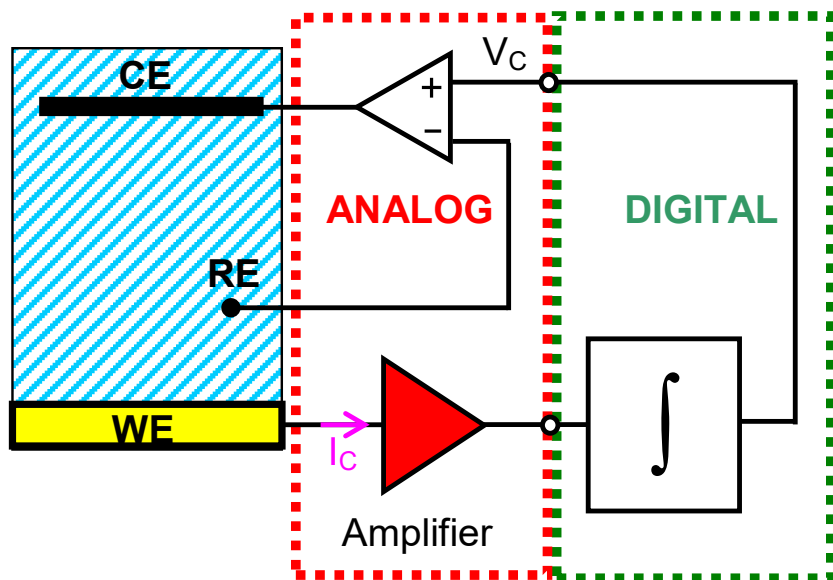


wide bandwidth more difficult (transimp. amp. in the feedback)

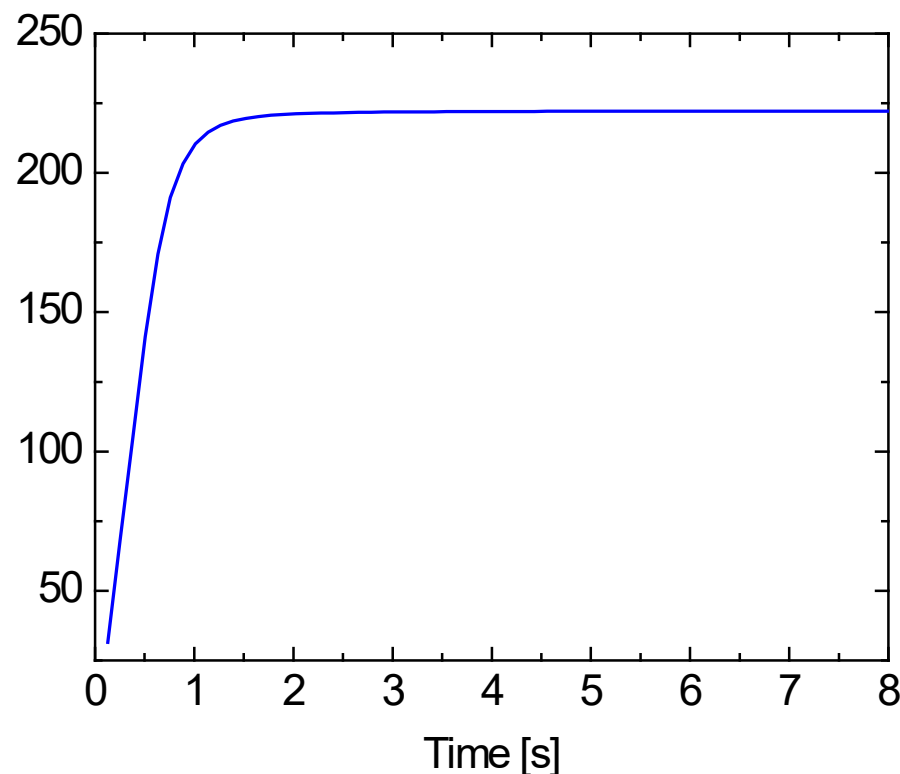


A particular galvanostatic case: current set-point = 0

Measure the equilibrium potential of the interface



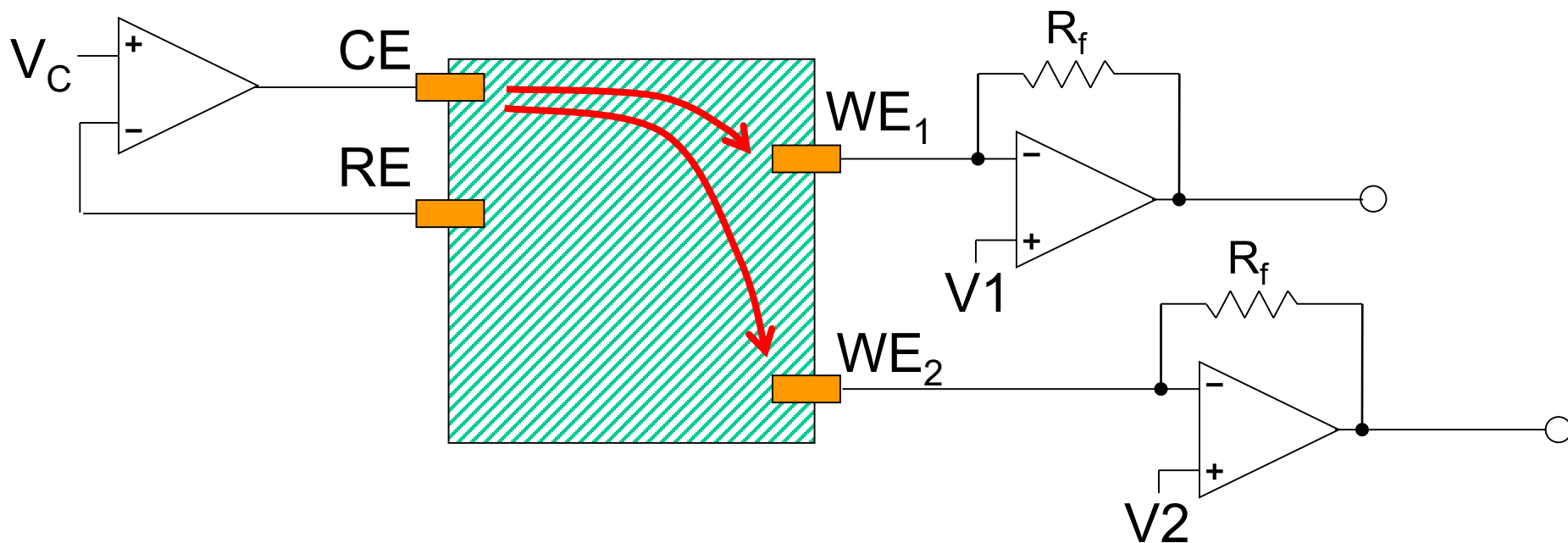
Potential [mV]



The loop must be stable

The Bipotentiostat Configuration

Multiple working electrodes (biased at different potentials) can be sensed in the same solution:



The single CE must handle the total current of all the WEs

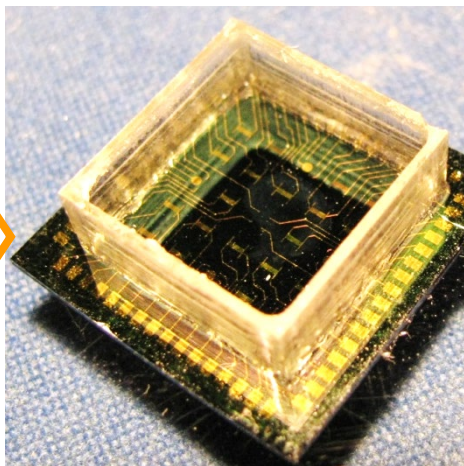
- The input current can be measured in several ways
 - Advanced circuit topologies can be adopted in order to improve the performance of the transimpedance amplifier
- Lez. Current Detection
- In order to measure impedance (in time tracking or frequency spectroscopy) a different circuit/processing architecture is needed (lock-in demodulator)
- Lez. Impedance
- Custom circuits, tailored for a specific application can perform better than general purpose potentiostats
 - Miniaturization, portability and integration have become fundamental design paradigms



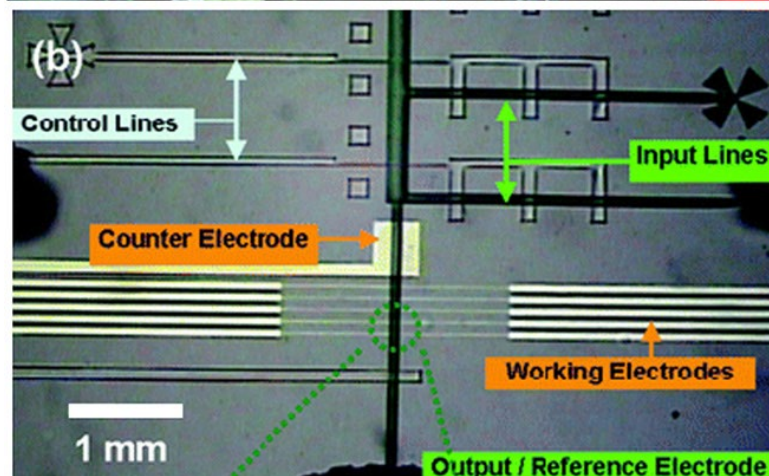
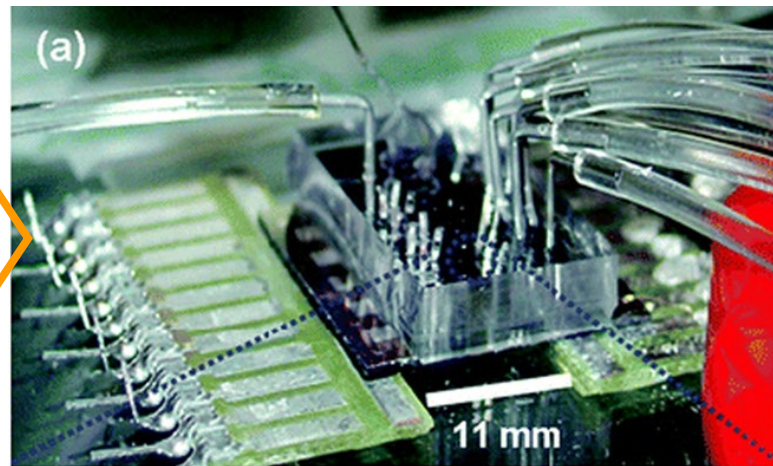
Echem. Detection Miniaturization Trend



Standard microscopic electrochemical cell



Static cell with microfabricated planar electrodes



Microelectrodes coupled with microfluidics

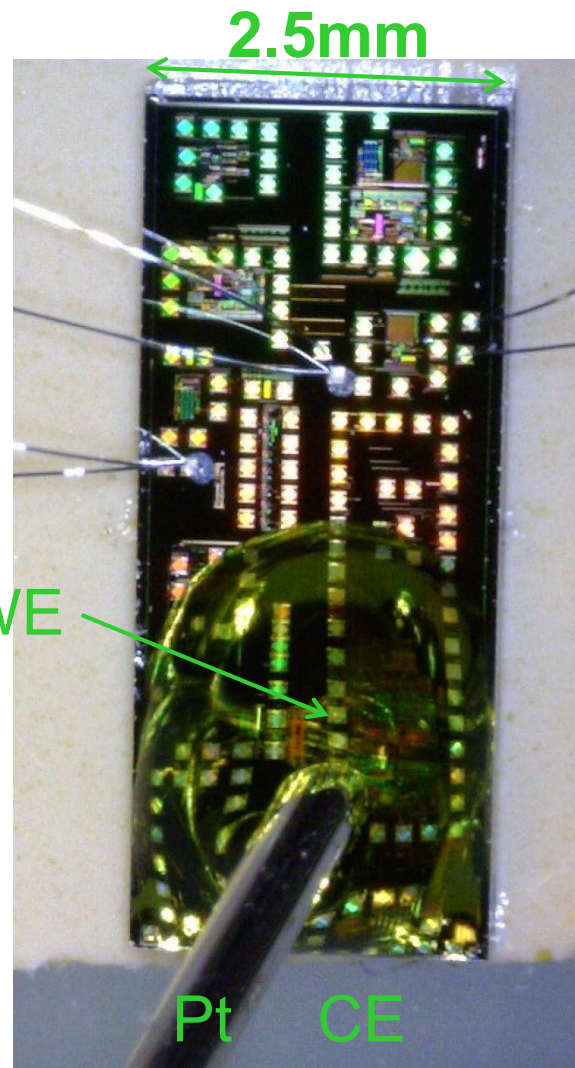


Shrinking the Potentiostat

Instrumentation should be miniaturized as well!



Standard bench-top instrument

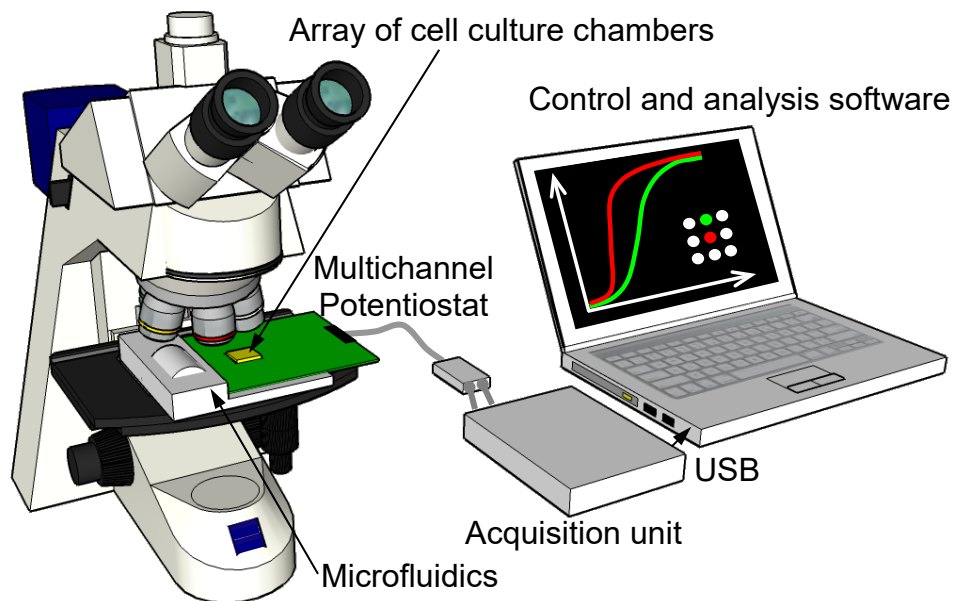


Miniaturization Advantages:

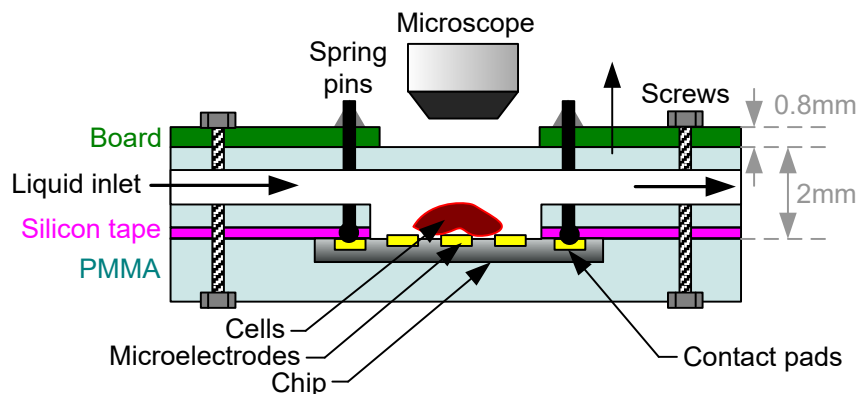
- Compactness (several channels in parallel)
- Performance (smaller input parasitics)



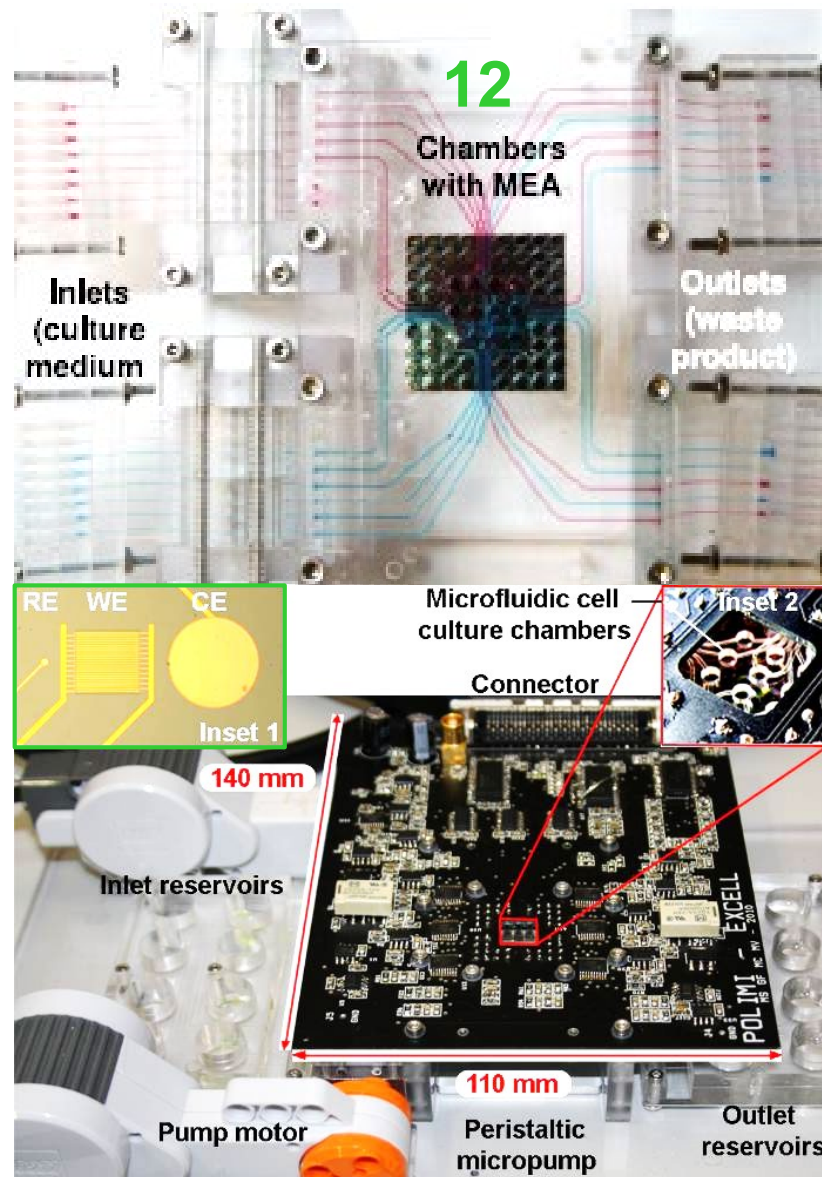
Electronics Integrated with the Setup



24 Channels, 100kHz BW

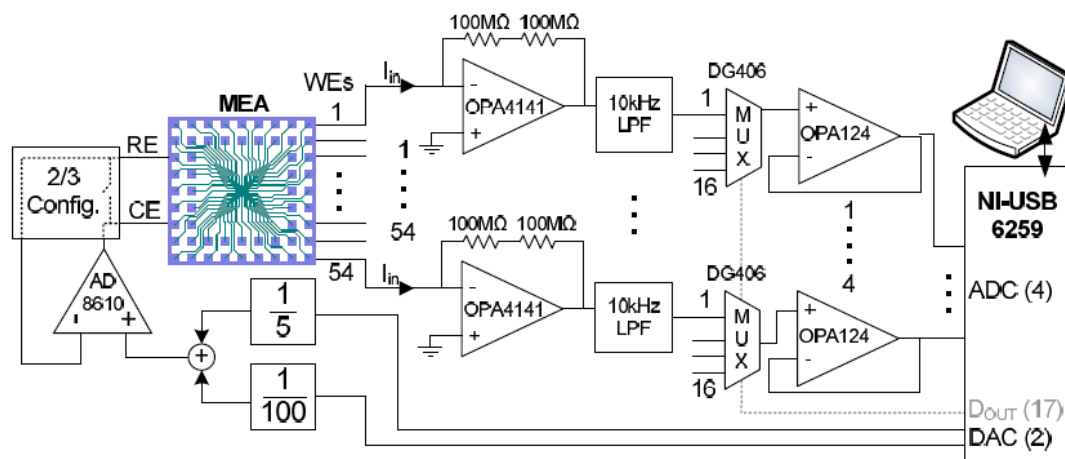
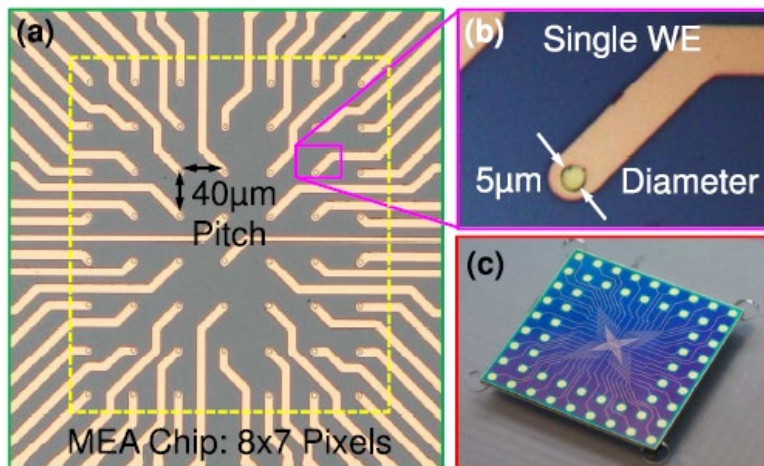


M. Vergani *et al.* TBCAS 2012

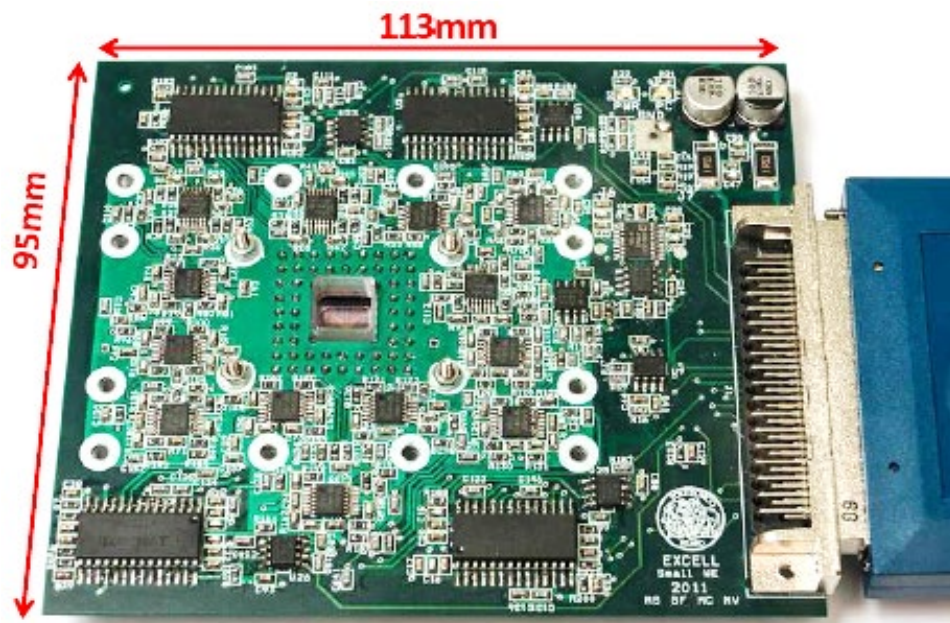




Example: 54 Channels



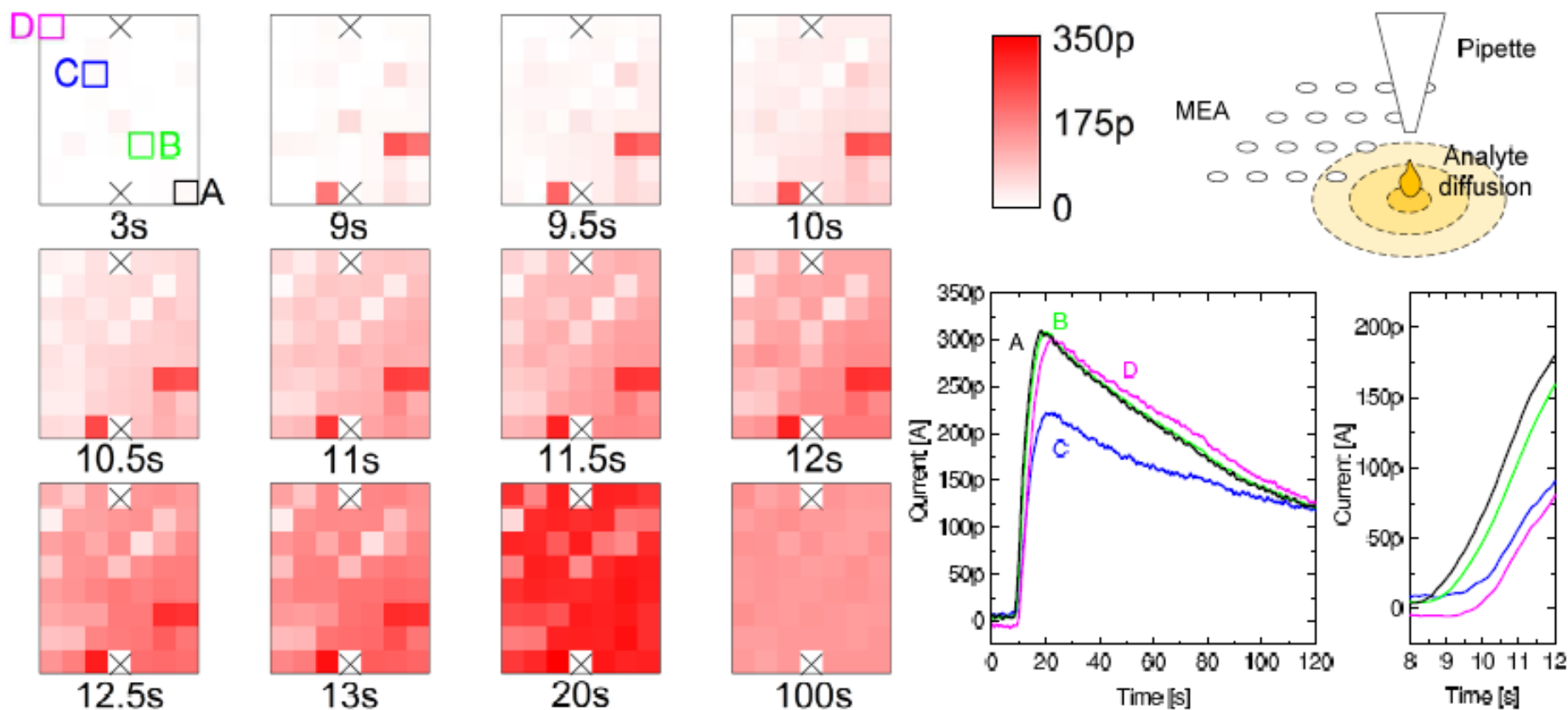
54 Channels, 8kHz BW
3pA resolution
Connectivity issues
Power issues: 3.8W





Example of Electrochemical Imaging

Tracking in real time the diffusion of the redox molecules:





Attofarad resolution potentiostat for electrochemical measurements on nanoscale biomolecular interfacial systems

Marco Carminati, Giorgio Ferrari, and Marco Sampietro

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498

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS, VOL. 6, NO. 5, OCTOBER 2012

Multichannel Bipotentiostat Integrated With a Microfluidic Platform for Electrochemical Real-Time Monitoring of Cell Cultures

Marco Vergani, Marco Carminati, *Member, IEEE*, Giorgio Ferrari, *Member, IEEE*, Ettore Landini, Claudia Caviglia, Arto Heiskanen, Clément Comminges, Kinga Zór, David Sabourin, Martin Dufva, Maria Dimaki, Roberto Raiteri, Ulla Wollenberger, Jenny Ennéus, and Marco Sampietro

Compact Potentiostat for Cellular Electrochemical Imaging with 54 Parallel Channels

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